



Non-isothermal consolidation: A systematic evaluation of two implementations based on multiphase and Richards equations

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ABSTRACT

In this paper, the governing equations of non-isothermal two-phase flow in unsaturated, deformable porous media are presented based on different representations of the gaseous phase along with their implementations in the open-source FEM code OpenGeoSys 6. As one implementation utilises the Richards equation to represent the unsaturated system whereas the second implementation relies on a two-component two-phase flow formulation, the impact of a free gaseous phase on model predictions is illustrated by a series of tests of increasing complexity, motivated by similar investigations in Task C of the DECOVALEX 2023 project. We demonstrate that the governing equations of both implementations collapse and become identical in fully saturated regions and that the numerical implementations remain stable in the special cases. Furthermore, the implicit assumptions made in the Richards equation are explicitly applied to the physically more complex and comprehensive TH²M model. Thus, the impact of a constant residual gas pressure on the model predictions is investigated. Finally, the examples chosen for this work feature thermal consolidation and pore fluid pressurisation effects in the Opalinus Clay due to heat release from a nuclear waste canister as well as dry-out with subsequent resaturation of a bentonite buffer around the heat source. All test cases are evaluated on the level of the governing equations of each model as well as on the level of results obtained by each model for a discussion of conceptual model uncertainty.

1. Introduction

In the context of the deep geological disposal of high-level radioactive waste (HLW), the usage of numerical tools is indispensable for the evaluation and prediction of the repository system behaviour over temporal and spatial scales beyond the reach of laboratory or in-situ experiments.¹ Examples for such numerical tools are codes, which are capable of numerically solving the partial differential equations (PDE) describing the thermo-hydro-mechanically (THM) coupled system behaviour in porous media and there exist syntheses in the literature presenting and comparing various examples of such codes.^{2–6} In deep geological HLW repositories involving clayey buffer materials or host rocks, typical THM-coupled processes include but are not restricted to thermal pressurisation of pore fluids in the host rock and generation of thermal stresses in the porous medium due to an increase of temperature,⁷ stress re-distribution as a consequence of excavation⁸

and desiccation⁹ and gas generation due to corrosion of metallic components in the Ref. 10. However, a numerical model can only describe a physical process as well as the underlying physical models and assumptions that are considered by the PDE in the first place. This gives rise to uncertainty with regard to the conceptual physical model, which differs from other types of uncertainty (e.g. regarding model parameters, model geometry as well as features of the spatial/temporal discretisation or numerical settings, which can be tackled by conducting ensemble analyses varying sensitive model parameters^{11–16}). If the uncertainties due to underlying physical assumptions of the model are themselves the object of investigation, an alternative strategy must be employed: The comparison of results obtained by different codes and conceptual models has proven to be a useful method to build confidence in the numerical tools and models and to increase process

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understanding,^{3,6,17} but it is not always clear whether differences observed in the model results are due to the set of PDEs, the numerical method or its implementation into software. A suitable solution pursued in this work is the comparison of two numerical models within the same code, one of which includes and one of which neglects a specific assumption in question (Fig. 1). Given the same temporal and spatial discretisation, the same numerical solver method and the same model parameterisation, this narrows down possible reasons for diverging results to said simplifying assumption. It can therefore help to provide evidence whether the assumption in question is justified for a specific application, and show how model predictions are affected by this choice.

The simplifying assumption discussed in this work is the approximation leading to the Richards Equation,¹⁸ which, coupled with temperature transport and mechanical processes, is implemented in OpenGeoSys 6 (OGS 6) and called the non-isothermal Richards mechanics (TRM) model¹⁹ henceforth. As a reference model, we use the non-isothermal two-phase two-component flow coupled with mechanics implementation (TH²M),²⁰ also available in the open-source FEM code OGS 6.²¹ The hydro-mechanical coupling is treated similarly in the two models and it is based on Biot's consolidation theory.^{22,23} Both the TRM and TH²M implementations represent two-phase flow in unsaturated, deformable media based on the concept of linear poro-elasticity, but in the case of the Richards approximation, the underlying physical simplification of this approximation is the assumption that the mobility of the gaseous phase is always high enough for gas drainage at a rate higher or faster than the other processes taking place. Therefore, it is assumed that there is no significant gas pressure build-up in the unsaturated porous medium and that the gas pressure remains constant.¹⁹ Despite this assumption, the Richards equation was extended to represent the process of water vaporisation and diffusive water transport (heat pipe effect)^{24,25} by allowing vapour pressure gradients and subsequent diffusive vapour transport in the gaseous phase. The TH²M model takes into account two phases (liquid and gaseous phase) and two components (water and non-aqueous component) which can each be present in both phases. Thus, phase transitions can be simulated and water vapour diffusion is taken into account as part of the more fundamental phase transition model. Since aside from water, other gas components may be present in a nuclear waste repository, state-dependent properties of this gas mixture will lead to a combination of diffusion and advection in the gaseous phase. Therefore, it is not trivial to quantify precisely how results produced by numerical models employing the Richards approximation are affected – a case of conceptual model uncertainty, which can be dealt with by applying a model of higher complexity explicitly considering the impact of a “free” gaseous phase, such as the TH²M model.

In this work, we evaluate and compare results obtained by the non-isothermal Richards mechanics (TRM) model and the non-isothermal two-phase two-component flow with mechanics (TH²M) model, both implemented in OGS 6.^{19–21} Studies comparing Richards and two phase flow codes for HM-coupled problems were performed by Ref. 4, and TH-coupled codes can consider models to account for mechanical effects to an extent,^{26–28} but comparison of the TRM and TH²M models including the fully coupled deformation process considered in the present work represents a new contribution to the literature.²⁹ demonstrated for isothermal problems that a non-constant pressure of the gaseous phase can have a significant impact on the calculated saturation and on the velocity at which a system reaches an equilibrium. This seems to be the case for systems in which gas and capillary pressures are in the same order of magnitudes and when gravity is the sole driving force. In the present work, however, the considered pressure ranges are much higher. A formulation for a fully coupled TH²M model was presented by Ref. 30, which represents in fact at least partly the basis for similar formulations in OGS,³¹ but it does not cover phase transitions like water evaporation.²⁰ Finally,³² shows how different

isothermal advective flow formulations compare and especially considers the simplifications of the governing equations of a tri-phasic model with regard to the two-phasic Richards model and chooses a similar numerical demonstration as Ref. 29. Studies comparing codes employing two-phase formulations and Richards approximations have been performed for TH-coupled problems,¹⁹ where it was shown that for a heat-pipe experiment the model predictions agree for media with relatively high intrinsic permeability, whereas in cases of low permeability, the Richards approximation did not predict experimental observations as accurately.¹⁹ also modelled a benchmark similar to the example discussed in this work, but results of the different models are discussed with a focus on the unsaturated bentonite backfill in the near-field of the heater. This work extends this comparison by also considering the thermal pressurisation in the fully saturated single-phase region in the far-field host rock, a process whose simulation requires the consideration of the mechanical process as well. By considering examples of decreasing complexity, we show that the more complex models collapse into simpler PDE formulations and that they do so as well in the simulations (cross-verification). Furthermore, we show how model complexity affects predictions using the example of the Richards approximation and the comparison to simulations with a “free” gaseous phase. Hence, conceptual model uncertainty tied to the Richards approximation with regard to gas-induced effects are addressed, using an application-oriented benchmark motivated by the Task C defined within the DECOVALEX 2023 project, an international code comparison and verification project.¹⁷ Lastly, we extend the work by Ref. 19 towards mechanical coupling. In the first section, the governing and constitutive equations of the two TRM and TH²M implementations are presented in a unified notation, thus allowing for a direct comparison. The second section deals with the balance equations under certain simplifying assumptions (namely the Richards assumption and single-phase flow assumption) and demonstrates how the PDEs collapse into simpler formulations. At all stages, the PDEs of the TH²M and TRM are compared, and the differences are discussed. The final section introduces the application case taken from the DECOVALEX 2023¹⁷ project, where 2D benchmarks motivated by the full-scale emplacement (FE) heater experiment at Mt. Terri³³ have been modelled. All the test cases discussed in the context of the governing PDEs are simulated using the TRM and TH²M implementations, and the results obtained by the TRM and TH²M models are compared.

2. TH²M and TRM implementations

Two new implementations available in the open-source framework OpenGeoSys 6²¹ are used in this study, and their governing and constitutive equations are given in this section. The first one is the non-isothermal two-phase two-component flow with mechanics model (TH²M).²⁰ The second model is the non-isothermal Richards mechanics (TRM) implementation, based on the non-isothermal Richards flow equation presented in the work by Ref. 19. Both models assume continuous, deformable porous media whose pore space is occupied by liquid and gaseous fluids. The TH²M model distinguishes itself from the TRM model in the use of a variable pressure for both fluid phases, each being comprised of two distinct components. As the TRM model considers water in the liquid phase and water vapour, it can be interpreted as a special simplified case of the TH²M model, in which the non-aqueous component is considered explicitly neither in the gaseous nor liquid phase. In the following chapters, the balance equations are given without explicit source or sink terms for simplicity. An overview of variables and abbreviations used in this work is given in Appendix C, Table 10.

Full liquid saturation	Gaseous phase presence (isobaric)	Non-isobaric gaseous phase
THM		
TRM-S	TRM	
TH ² M-S	TH ² M-R	TH ² M

Fig. 1. Process diagram illustrating single and two phase flow regions along with their coverage by the different process models discussed in this paper. The goals are consistency across process model implementations on the one side and “downward compatibility” within complex process models when transitioning to simpler physical settings on the other.

2.1. Mass balance equations

2.1.1. TH²M mass balance

In the TH²M implementation presented in Ref. 20, a liquid and a gaseous phase, $\alpha \in \{L, G\}$, are considered which occupy the pore space of a deformable porous solid $\alpha \equiv S$. Each fluid phase is populated by the constituents/components ζ . In our case, one of the components is always water, while the other remains unspecified at this point, i.e. $\zeta \in \{W, C\}$. The coupled processes of energy transport, deformation and two-phase two-component hydraulics are solved using four balance equations with temperature T , solid displacement vector $\mathbf{u}_S$, gaseous phase pressure p_{GR} and capillary pressure p_{cap} as primary variables.^e The liquid pressure is expressed in terms of the primary variables:

$$p_{LR} = p_{GR} - p_{cap}. \quad (1)$$

In order to model phase transitions between the two fluid phases (namely evaporation and condensation of the water component into and out of the gaseous phase as well as dissolution and out-gassing of the complementary component into and out of the liquid phase), the two mass balance equations are written in a component form. Instant phase equilibration is assumed so that the two-phase two-component formulation requires no additional degree of freedom to describe phase change. The component mass balance equations read:

$$\begin{aligned} 0 = & \rho_{FR}^{\zeta} (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{GR}}{dt} - \rho_{FR}^{\zeta} (\alpha_B - \phi) \beta_{p,SR} S_L \frac{dp_{cap}}{dt} \\ & - \rho_{FR}^{\zeta} (\alpha_B - \phi) \text{tr}(\alpha_{T,SR}) \frac{dT}{dt} \\ & + \phi \left[(1 - S_L) \frac{d\rho_{GR}^{\zeta}}{dt} + S_L \frac{d\rho_{LR}^{\zeta}}{dt} \right] \\ & + \left[\phi (\rho_{LR}^{\zeta} - \rho_{GR}^{\zeta}) - \rho_{FR}^{\zeta} p_{cap} (\alpha_B - \phi) \beta_{p,SR} \right] \frac{dS_L}{dt} \\ & + \rho_{FR}^{\zeta} \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div}(\mathbf{A}^{\zeta} + \mathbf{J}^{\zeta}), \end{aligned} \quad (2)$$

where $\frac{d\circ}{dt}$ is the material time derivative following the velocity of the moving solid, α_B and ϕ represent the Biot–Willis coefficient and porosity, respectively, $\beta_{p,SR}$ is the solid grain compressibility, $\alpha_{T,SR}$ is a diagonal matrix for the linear solid thermal expansion coefficients in all coordinate directions such that $\text{tr}(\alpha_{T,SR})$ gives the volume thermal expansion coefficient of the solid, and the effective density of component ζ in both fluid phases is given by:

$$\rho_{FR}^{\zeta} = (1 - S_L) \rho_{GR}^{\zeta} + S_L \rho_{LR}^{\zeta} \quad (3)$$

^e Note regarding the subscript ‘R’: The theory of porous media describes the interaction between substitute continua which appear by upscaling quantities defined on the level of the constituting materials from the pore scale to the continuum scale of the porous medium using appropriate averaging procedures. This leads to the definition of apparent properties of the substitute continua linked to the intrinsic (or ‘real’, hence the subscript ‘R’) properties of the constituting materials by means of volume fractions. In the case of hierarchical porous media, these intrinsic properties themselves could be the result of an upscaling from yet a lower scale, but this case it not discussed here.

with ρ_{GR}^{ζ} and ρ_{LR}^{ζ} as the density of component ζ in each respective phase. The component densities are defined via the phase densities: The liquid phase density ρ_{LR} can be given by a multilinear function²⁰:

$$\rho_{LR} = \rho_{LR,ref} (1 + \beta_{p,LR} (p_{LR} - p_{LR,ref}) - \beta_{T,LR} (T - T_{ref}) + \beta_{c,LR} c_L^C) \quad (4)$$

with the reference values $\rho_{LR,ref}$, $p_{LR,ref}$ and T_{ref} and with $\beta_{c,LR}$ as the liquid expansivity due to the amount of dissolved gas c_L^C defined in Eq. (15). $\beta_{p,LR}$ and $\beta_{T,LR}$ represent the liquid compressibility and thermal expansion. Although in this work gas dissolution is prevented and hence, $c_L^C = 0$ in Eq. (4), we proceed to summarise the component densities in the liquid for the sake of completeness: The component densities in the liquid phase can then be calculated using Henry’s law Eq. (14) for ρ_{LR}^C (gas component) and using $\rho_{LR}^W = \rho_{LR} - \rho_{LR}^C$ for the water component. The gaseous phase density is given by applying the ideal gas law:

$$\rho_{GR} = \frac{p_{GR} M_G}{RT}, \quad (5)$$

where R is the universal gas constant and M_G is the effective gas phase molar mass, defined just below Eq. (9). The density of each component ζ is subsequently given by $\rho_{GR}^{\zeta} = x_{m,G}^{\zeta} \rho_{GR}$, where the mass fraction $x_{m,G}^{\zeta}$ is defined in Eq. (9). The liquid saturation S_L is a function of capillary pressure p_{cap} , in this work expressed according to van Genuchten with $S_L = S_e (S_{L,max} - S_{L,res}) + S_{L,res}$ where the effective saturation S_e is given according to Ref. 34:

$$S_e = \left(1 + \left(\frac{p_{cap}}{p_e} \right)^n \right)^{\frac{1}{n-1}}, \quad (6)$$

where p_e represents the gas entry pressure and $n \in [0, 1)$ is an empirical parameter. The dispersive–diffusive mass fluxes $\mathbf{J}^{\zeta} = \mathbf{J}_L^{\zeta} + \mathbf{J}_G^{\zeta}$ of component ζ in both fluid phases are driven by the constituent concentration gradient in each respective phase, which – due to the phase equilibrium assumption – is a function of gas pressure, capillary pressure and temperature (cf. Eqs. (10), (11) and (12), respectively): Consequently, the diffusive-dispersive mass flux is expressed by

$$\begin{aligned} \mathbf{J}_G^{\zeta} = & -\rho_{GR} \mathbf{D}_G^{\zeta} \text{grad } x_{m,G}^{\zeta} \\ = & -\rho_{GR} \mathbf{D}_G^{\zeta} \left(\frac{\partial x_{m,G}^{\zeta}}{\partial p_{GR}} \text{grad } p_{GR} + \frac{\partial x_{m,G}^{\zeta}}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial x_{m,G}^{\zeta}}{\partial T} \text{grad } T \right) \end{aligned} \quad (7)$$

for the gaseous phase and

$$\begin{aligned} \mathbf{J}_L^{\zeta} = & -\rho_{LR} \mathbf{D}_L^{\zeta} \text{grad } x_{m,L}^{\zeta} = -\rho_{LR} \mathbf{D}_L^{\zeta} \left(\frac{\partial x_{m,L}^{\zeta}}{\partial p_{LR}} (\text{grad } p_{GR} - \text{grad } p_{cap}) \right. \\ & \left. + \frac{\partial x_{m,L}^{\zeta}}{\partial p_{GR}} \text{grad } p_{GR} + \frac{\partial x_{m,L}^{\zeta}}{\partial T} \text{grad } T \right) \end{aligned} \quad (8)$$

for the liquid phase, where $\mathbf{D}_G^{\zeta} = D_G^{\zeta} \phi (1 - S_L) \mathbf{I}$ and $\mathbf{D}_L^{\zeta} = D_L^{\zeta} \phi S_L \mathbf{I}$ represent the isotropic diffusivity tensors. The diffusive transport of each component in the liquid and gaseous phase is driven by its mass fraction gradient in the respective fluid phase. In Eqs. (9)–(12), the component mass fractions in the gaseous phase are defined and in Eqs. (13)–(15), the component mass fractions in the liquid phase are defined. The mass fraction of each component in the gaseous phase is related to its molar fraction via the molar mass ratios:

$$x_{m,G}^{\zeta} = x_{n,G}^{\zeta} \frac{M_{\zeta}}{M_G}, \quad (9)$$

which is in turn defined by the ratio of water vapour pressure and gas pressure, Eq. (10). M_ζ represents the molar mass of component ζ and $M_G = \sum_\zeta x_{n,G}^\zeta M_\zeta$ represents the effective gaseous phase molar mass. Dalton's law, which states that the gaseous phase pressure equals the sum of partial pressures of each component in the gaseous phase, is applied to define the molar fraction of the non-aqueous component in the gaseous phase:

$$x_{n,G}^W = \frac{p_{GR}^W}{p_{GR}} \text{ and } x_{n,G}^C = 1 - x_{n,G}^W. \quad (10)$$

The water vapour pressure corrected for phase-interface curvature according to Kelvin–Laplace is given by

$$p_{GR}^W = p_{vap}^W \exp\left(-\frac{p_{cap} M_W}{\rho_{LR} RT}\right), \quad (11)$$

where R is the universal gas constant and M_W is the molar mass of water. As opposed to the constitutive laws presented by Ref. 20, the saturated vapour pressure p_{vap}^W is calculated in this work by a different empirical law. It was chosen so that it yields a saturated vapour pressure, which is equivalent to the saturated vapour density calculated for the TRM model (cf. Eq. (21)):

$$p_{vap}^W = \frac{RT}{M_W} \rho_{vap}^W, \quad (12)$$

where ρ_{vap}^W is the saturated vapour density given in Eq. (21). The diffusive transport in the liquid phase as defined in Eq. (8) is driven by the mass fraction gradient of component ζ in the liquid phase, which is given by:

$$x_{m,L}^C = \frac{\rho_{LR}^C}{\rho_{LR}}. \quad (13)$$

The density of the dissolved non-aqueous component in the liquid phase is proportional to its partial pressure in the gaseous phase and is given by Henry's law:

$$\rho_{LR}^C = \rho_{LR,ref} \beta_{c,LR} c_L^C, \quad (14)$$

$$c_L^C = H^C p_{GR}^C = H^C x_{n,G}^C p_{GR} \quad (15)$$

with a Henry's coefficient H^C which can be assumed to be constant or temperature-dependent.^{20,35} Finally, the water component mass fraction in the liquid is given by $x_{m,L}^W = 1 - x_{m,L}^C$. The effective advective mass flux of component ζ is defined by its advective mass flux in each fluid phase $\mathbf{A}^\zeta = \mathbf{A}_L^\zeta + \mathbf{A}_G^\zeta$. In each respective fluid phase, $\alpha \in \{L, G\}$, the advective mass flux is given by the Darcy velocity $\tilde{\mathbf{w}}_{\alpha S}$ and the partial density of the component in the respective phase ρ_α^ζ :

$$\mathbf{A}_\alpha^\zeta = \rho_{\alpha R}^\zeta \tilde{\mathbf{w}}_{\alpha S} = -\rho_{\alpha R}^\zeta \frac{k_\alpha^{\text{rel}} k_S}{\mu_{\alpha R}} (\text{grad } p_{\alpha R} - \rho_{\alpha R} \mathbf{g}), \quad (16)$$

where k_S represents the anisotropic intrinsic medium permeability tensor, while in this work, the liquid viscosity is given by the empirical relation³⁶

$$\mu_{LR} = 4.28 \times 10^{-5} \text{ Pa s} + \frac{1 \text{ Pa s}}{0.16 \cdot \left(\frac{T}{K} + 338.15\right)^2 - 91.3}, \quad (17)$$

the gaseous phase viscosity μ_{GR} and relative permeability k_G^{rel} are assumed to be constants, and the liquid relative permeability is defined according to van Genuchten/Mualem³⁷:

$$k_L^{\text{rel}} = \sqrt{S_e} \left(1 - \left(1 - S_e^{n/(n-1)}\right)\right)^{(2n-2)/n}. \quad (18)$$

The body force vector $\mathbf{g}$ represents the earth's gravitational acceleration acting on all phases. However in this work, gravitation is neglected in the numerical examples for simplicity.

2.1.2. TRM mass balance

In the TRM implementation,¹⁹ the water mass balance is formulated in terms of liquid pressure p_{LR} ; similarly to the TH²M implementation, the other main variables are the temperature T as well as the solid displacement vector $\mathbf{u}_S$. Water vapour flux in the TRM model is realised using the vapour diffusion concept presented by Ref. 24 and the presence of water vapour is taken into account in the thermal expansion term and in the storage-related terms. The liquid mass balance equation reads:

$$\begin{aligned} 0 = & \rho_{LR} S_L^2 (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{LR}}{dt} - \rho_{LR} S_L (\alpha_B - \phi) \text{tr}(\alpha_{T,SR}) \frac{dT}{dt} \\ & + \phi \left((1 - S_L) \frac{d\rho_{GR}^W}{dt} + S_L \frac{d\rho_{LR}}{dt} \right) + (\rho_{LR} - \rho_{GR}^W) \\ & \times [\phi + p_{LR} S_L (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt} \\ & + S_L \rho_{LR} \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div}(\mathbf{A}_L^W + \mathbf{J}_G^W), \end{aligned} \quad (19)$$

where ρ_{LR} is the liquid phase density, which is calculated using Eq. (4) assuming that $c_L^C = 0$ by definition. The advective flux $\mathbf{A}_L^W$ is given by Eq. (16) and the liquid saturation S_L is given by Eq. (6), where $p_{cap} = -p_{LR}$ is assumed. Although water vapour-related effects are taken into account, there is no additional degree of freedom for the water vapour, i.e. no additional balance equation is used to describe the vapour/gaseous phase. Instead, the formulas of its properties are derived from the thermal and hydraulic conditions using the following constitutive equations. The vapour diffusion is expressed in terms of vapour densities as opposed to the TH²M implementation, where it was expressed in terms of vapour pressure or component mass fractions (cf. Eqs. (7)–(12)). The vapour density corrected for phase-interface curvature according to Kelvin–Laplace is given by:

$$\rho_{GR}^W = \rho_{vap}^W \exp\left(-\frac{p_{LR}}{\rho_{LR} RT}\right), \quad (20)$$

where the saturated water vapour density ρ_{vap}^W is given by an empirical relation³⁸ resulting in equivalence to what is defined for the TH²M model in Eq. (12):

$$\rho_{vap}^W = 10^{-3} \frac{\text{kg}}{\text{m}^3} \cdot \exp(19.819 - 4975.9 \text{ K}/T). \quad (21)$$

The diffusion or vapour flux is given by:

$$\mathbf{J}_G^W = -\mathbf{D}_G^W \text{grad } \rho_{GR}^W = -\mathbf{D}_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{LR}} \text{grad } p_{LR} + \frac{\partial \rho_{GR}^W}{\partial T} \text{grad } T \right). \quad (22)$$

The diffusion coefficient can be defined by various models. In order to compare the TRM and TH²M models consistently, a definition of vapour diffusivity in air is given by $\mathbf{D}_G^W = D_G^\zeta \phi (1 - S_L) \mathbf{I}$.^{19,20,39} When comparing the mass balance equations in the two implementations, some similarities can be observed: Both models consider medium deformation, thermal expansivity of the liquid and solid phase, saturation changes and so on. These similarities are discussed in detail in the following section. While the diffusive transport of water vapour is expressed in terms of vapour density gradients in TRM (cf. Eq. (22)) and vapour mass fractions in TH²M (cf. Eq. (7)), the constitutive laws establishing the saturated vapour properties (Eqs. (21) and (12) for TRM and TH²M, respectively) and effective vapour properties (Eqs. (20) and (11) for TRM and TH²M) are equivalent and can be converted using the ideal gas law.

2.2. Momentum balance equations

In both the TH²M and TRM implementations, the momentum balance equation follows Biot's effective stress theory. They differ only with regard to the consideration of pore gas pressure and pore gas density.

$$\mathbf{0} = \begin{cases} \text{TH}^2\text{M} : & \text{div}(\boldsymbol{\sigma}' - \alpha_B [\chi p_{LR} + (1 - \chi) p_{GR}] \mathbf{I}) \\ & + [(1 - \phi) \rho_{SR} + S_L \phi \rho_{LR} + (1 - S_L) \phi \rho_{GR}] \mathbf{g} \\ \text{TRM} : & \text{div}(\boldsymbol{\sigma}' - \alpha_B \chi p_{LR} \mathbf{I}) + [(1 - \phi) \rho_{SR} + S_L \phi \rho_{LR}] \mathbf{g}, \end{cases} \quad (23)$$

where $\sigma' = \mathbf{C} : (\epsilon - \epsilon_{\text{inelastic}} - \epsilon_{\text{thermal}})$ represents the effective stress tensor with the thermal strain $\frac{d}{dt}\epsilon_{\text{thermal}} = \alpha_{T,SR} \frac{dT}{dt}$ and $\mathbf{I}$ representing the identity. χ is the Bishop coefficient, which is typically chosen to be saturation-dependent. For the purposes of this paper, we simplify matters by assuming $\chi = S_L$. Another common practice is a cut-off value with

$$\chi = \begin{cases} 0, & \text{if } S_L \leq S_{\text{entry}} \\ 1, & \text{if } S_L > S_{\text{entry}} \end{cases} \quad (24)$$

where S_{entry} is the saturation at gas entry.

2.3. Energy balance equations

2.3.1. TH²M enthalpy balance

The enthalpy balance in the TH²M implementation is given by Ref. 20:

$$\begin{aligned} 0 = & \underbrace{\frac{d}{dt}(\Sigma_\alpha \rho_\alpha u_\alpha)}_{\text{storage}} + \underbrace{\text{div}(\Sigma_\alpha h_\alpha \mathbf{A}_\alpha)}_{\text{transport by advection}} \\ & + \underbrace{\text{div}(\Sigma_\alpha \Sigma_\zeta h_\alpha^\zeta \mathbf{J}_\alpha^\zeta)}_{\text{transport by diffusion}} + \underbrace{(\Sigma_\alpha \rho_\alpha h_\alpha) \text{div}\left(\frac{d\mathbf{u}_S}{dt}\right)}_{\text{medium volume change}} - \underbrace{\Sigma_\alpha \mathbf{g} \cdot \mathbf{A}_\alpha}_{\text{gravitation work}} \\ & - \underbrace{\text{div}(\lambda_{\text{eff}} \text{grad } T)}_{\text{heat conduction}}, \end{aligned} \quad (25)$$

where ρ_α is the apparent density of a phase which, expressed for each phase, reads $\rho_L = \phi S_L \rho_{LR}$, $\rho_G = \phi(1 - S_L) \rho_{GR}$ and $\rho_S = (1 - \phi) \rho_{SR}$, respectively. u_α is the specific internal energy of each phase and $h_\alpha = \frac{p_\alpha}{\rho_\alpha} + u_\alpha$ is the specific enthalpy. From the definitions of u_α and h_α , one can observe that pressure-work is included in the formulation. The advective mass flux of α represents the phase mixtures containing both components, and it is given by $\mathbf{A}_\alpha = \rho_{\alpha R} \tilde{\mathbf{w}}_{\alpha S}$ with the Darcy velocity defined according to Eq. (16). Details of the constitutive laws can be found in Refs. 20, 40, 41. λ_{eff} is the effective medium thermal conductivity tensor, which can be calculated by different mixing laws – in this study, it is given by a saturation-dependent relation:

$$\lambda_{\text{eff}} = \lambda_{\text{dry}} + S_L (\lambda_{\text{sat}} - \lambda_{\text{dry}}). \quad (26)$$

The effective medium thermal heat capacity based on the liquid, gas and solid phase properties are averaged using the common mixing rule

$$(\rho c_p)_{\text{eff}} = \phi [(1 - S_L) \rho_{GR} c_{pG} + S_L \rho_{LR} c_{pL}] + (1 - \phi) c_{pS} \rho_{SR}, \quad (27)$$

where the phase heat capacities are in turn calculated using a mixing rule according to the phase compositions $c_{p\alpha} = \Sigma_\alpha x_{m,\alpha}^\zeta c_{p\zeta}$. The storage term in Eq. (25) can be expanded, and we will show later (cf. Eq. (33)) that it reflects both enthalpy stored due to temperature changes and enthalpy stored due to phase pressure changes and phase transitions. However, the numerical implementation of this storage term evaluates the change with respect to time of $\Sigma_\alpha \rho_\alpha u_\alpha$ directly (secant formulation)²⁰ and therefore, we use the present equation.

2.3.2. TRM energy balance

In the TRM implementation, the conservation of energy is given by:

$$\begin{aligned} 0 = & (\rho c_p)_{\text{eff}} \frac{dT}{dt} + L_0 \frac{d\theta_{\text{vap}}}{dt} - \text{div}(\lambda_{\text{eff}} \text{grad } T) + \text{div}\left(\frac{L_0 \mathbf{J}_G^W}{\rho_{LR}}\right) \\ & + \text{grad } T \cdot (c_{pL} \mathbf{A}_L + c_{p,\text{vap}} \mathbf{J}_G^W). \end{aligned} \quad (28)$$

This formulation can be derived from the general balance Eq. (25) by substituting the balance equations for mass and momentum as well as expanding time derivatives using the constitutive definitions of specific enthalpy etc. This gives rise to the explicit appearance of specific heat capacities, quantities related to the phase change, etc. As an additional simplification, the effects of pressure work/stress powers

have been neglected. L_0 represents the volumetric latent heat of water vaporisation,⁴² given in J m^{-3} :

$$L_0 = \rho_{LR} \left(2.501 \times 10^6 \text{ J kg}^{-1} - 2369.2 \text{ J kg}^{-1} \left(\frac{T}{K} - 273.15 \right) \right). \quad (29)$$

The effective thermal conductivity and the effective heat capacity of the medium can be calculated using the mixing rule in Eqs. (26) and (27), respectively. However, the water vapour contributions to the effective thermal heat capacity in Eq. (27) have been neglected in the TRM implementation. The volumetric vapour content is defined by Ref. 43:

$$\theta_{\text{vap}} = \phi(1 - S_L) \frac{\rho_{GR}^W}{\rho_{LR}}. \quad (30)$$

3. Case studies

The aim of this chapter is to present some differences between the two aforementioned model implementations by applying specific assumptions of three examples to the governing equations of the TH²M and TRM models. The three examples comprise

- a “general case”, where the full TRM model without simplification is used. In the TH²M model, we prevent the dissolution of the non-aqueous component into the liquid phase;
- a “Richards case”, where again the full TRM model without simplification is used. However, the assumptions of the Richards approximation intrinsic to the TRM model are applied explicitly to the TH²M model, yielding the TH²M-R model;
- a “single phase case”, where the assumption of full saturation is applied to both models, yielding the TRM-S and the TH²M-S models. As gas dissolution is neglected in the TH²M-S model, the system reduces to a single-component, single-phase problem.

Using this set of examples with increasing degrees of simplification, we analyse similarities and differences of the TRM and TH²M implementations. Furthermore, we study the impact of freely evolving gas pressures. Thus, the conceptual uncertainty related to the gaseous phase inherent in the TRM model will also be discussed. Later on, the numerical results corresponding to each of the three examples will be discussed in reverse order with increasing complexity.

3.1. General case: Full TH²M and TRM models without simplifying assumptions

In this general case, taking into account a freely evolving gas pressure and thus an additional degree of freedom, the TRM model does not necessarily represent the entire physical system accurately since it only considers the water mass balance and assumes that the gaseous phase is highly mobile and that there is no gas pressure build-up. The only simplification applied to the TH²M model is that no dissolution of the non-aqueous component into the liquid phase takes place. In order to compare the TH²M water component mass balance to the TRM water mass balance, Eq. (2) is evaluated for the water component with $\zeta \equiv W$. Since there is no dissolved gas in the liquid phase, the molar fraction of the water in the liquid phase $x_{m,L}^W = 1$ is constant (with regard to space and time) and therefore, its derivatives are equal to zero. Thus, there is no diffusive flux of the water component in the liquid phase ($\mathbf{J}^W = \mathbf{J}_L^W + \mathbf{J}_G^W = \mathbf{J}_G^W$). For the same reason, we simplify $\rho_{LR}^\zeta = \rho_{LR}^W = \rho_{LR}$ in Eq. (3). After expanding the density derivative terms, some reordering with respect to the driving forces and inserting the definition of the effective component density Eq. (3), we can express the water component mass balance in TH²M by (see Appendix A for

details):

$$\begin{aligned}
0 = & \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{GR}} + \phi S_L \rho_{LR} \beta_{p,LR} \right. \\
& + \left. ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} \right) \frac{dp_{GR}}{dt} \\
& + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{cap}} - \phi S_L \rho_{LR} \beta_{p,LR} \right. \\
& - \left. ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} S_L \right) \frac{dp_{cap}}{dt} \\
& + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \phi S_L \rho_{LR} \beta_{T,LR} \right. \\
& - \left. ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) (\alpha_B - \phi) \text{tr}(\alpha_{T,SR}) \right) \frac{dT}{dt} \\
& + ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div}(\mathbf{A}^W + \mathbf{J}^W) \\
& + [\phi(\rho_{LR} - \rho_{GR}^W) - ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) p_{cap} (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}.
\end{aligned} \quad (31)$$

Similarly to the treatment of the water component mass balance of the TH²M implementation described in Appendix A, we expand the derivatives[†] of the liquid and vapour densities in the TRM water mass balance Eq. (19) and reorder the equation with respect to the driving forces:

$$\begin{aligned}
0 = & \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{LR}} + \rho_{LR} S_L [\phi \beta_{p,LR} + S_L (\alpha_B - \phi) \beta_{p,SR}] \right) \frac{dp_{LR}}{dt} \\
& + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \rho_{LR} S_L [\phi \beta_{T,LR} + (\alpha_B - \phi) \text{tr}(\alpha_{T,SR})] \right) \frac{dT}{dt} \\
& + S_L \rho_{LR} \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div}(\mathbf{A}_L^W + \mathbf{J}_G^W) + (\rho_{LR} - \rho_{GR}^W) \\
& \times [\phi + p_{LR} S_L (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}.
\end{aligned} \quad (32)$$

After applying the definition of the advective and diffusive-dispersive transport of component $\zeta \equiv W$, using Eqs. (16), (7) and (8), we can now compare the water (component) balance equations of the TH²M and TRM models and assign a physical process to each term listed in Table 2. For the comparison of the energy balance equations in Table 3, we consider firstly the energy balance Eq. (25) implemented in the TH²M model. Since gas dissolution is prevented, diffusion occurs only in the gaseous phase and $\text{div}(\Sigma_\alpha \Sigma_\zeta h_\alpha^\zeta \mathbf{J}_\alpha^\zeta)$ simplifies to $\text{div}(\Sigma_\zeta h_G^\zeta \mathbf{J}_G^\zeta)$. Furthermore, we can expand the sums over the phases/components in order to compare the energy balance implementations of TH²M and TRM in Table 3. It can be seen therein that both implementations take into account energy storage, heat conduction and the advective heat transport, as well as some latent heat terms related to phase transitions. The TH²M model additionally takes into account energy stored in phase pressures of the phases (including work done by gravitation), which the TRM model ignores.

In summary, the comparison of the balance equations in the general example with a freely evolving gas pressure shows differences between the TRM and TH²M models in all three thermo-hydraulic and mechanic processes. When comparing the hydraulic process between TRM and TH²M, it firstly has to be stated that we only look at the water component mass balances. An obvious difference is that there is an additional mass balance (non-aqueous component mass balance) in TH²M, which is missing in TRM. Therefore, no direct comparison of the mass balances of the non-aqueous components can be made, instead the effects of a non-constant gas pressure on the water mass balance can be discussed.

Table 2 shows the direct comparison of the TRM and TH²M water component mass balances, and it can be seen that equivalent physical processes can be assigned to each term. Thus, we can discuss differences between TRM and TH²M for each process marked in Table 2:

- **Deformation.** The contribution of the medium deformation is equivalent in the two models. A difference arises only from the effective water component density in TH²M. The difference decreases with increasing saturation. However, since the deformation represents a process coupling to the mechanics, $\text{div}(\frac{d\mathbf{u}_S}{dt})$ is not necessarily equal, e.g. when high gas injection pressures cause a medium deformation.
- **Advection.** In addition to advective transport of the water component in the liquid phase, there is advective transport of vapour in the gaseous phase in TH²M. While in TRM, the liquid pressure governing the Darcy flux is a main variable, it is a secondary variable expressed by p_{GR} and p_{cap} in TH²M.
- **Diffusion.** In TRM, the vapour diffusion is driven by gradients in liquid pressure and temperature and is expressed in terms of vapour density. In TH²M, it is additionally driven by the gas pressure gradient, and it is expressed in terms of the water component mass fraction in the gaseous phase. The conversion from vapour density to mass fraction is discussed in the next chapter, and the impact of gas pressure gradients on the diffusion are discussed in the “general case” part in Section 4.4. It should be noted that if the dissolution of the non-aqueous component was allowed, there would be an additional concentration gradient with $\text{grad} x_{m,L}^W = -\text{grad} x_{m,L}^C$ driving diffusion of the water component in the liquid phase.
- **Thermal expansion.** The expressions in the two models approach each other for low vapour densities and high liquid saturations.
- **Storage due to partial saturation.** It can be seen that in TRM one would have to substitute $p_{LR} = -p_{cap}$ in order to achieve complete equivalence. This shows that the liquid-pressure-offset by the constant gas pressure (Eq. (1)) is neglected in this term in TRM. Apart from that, the expressions in the two models approach for high saturations or vanishing vapour density.
- **Storage due to liquid, gaseous or solid phase compression as a response to liquid pressure changes.** $\frac{dp_{cap}}{dt} = -\frac{dp_{LR}}{dt}$ and $\frac{\partial \rho_{GR}^W}{\partial p_{cap}} = -\frac{\partial \rho_{GR}^W}{\partial p_{LR}}$ can be substituted, and the expressions in the two models approach each other for high saturation or vanishing vapour density.
- **Storage due to liquid, gaseous or solid phase compression as a response to gas pressure changes.** This effect is neglected by definition in TRM. It is considered in TH²M and its formulation is analogous to compression due to liquid pressure changes.

When comparing the energy balance equations in Table 3, it can be seen that all terms can be assigned to equivalent processes. Heat conduction is expressed identically in both implementations, but all other processes differ in their implementation. Both models consider liquid phase flow for advective heat transport but in TH²M, the advection in the gaseous phase is additionally present. The TRM model uses the diffusive vapour flux in order to express advective heat and latent heat transport via water vapour flux. The energy storage in TH²M can be expanded by considering that $u_\alpha = h_\alpha - \frac{p_\alpha}{\rho_\alpha}$ and $\frac{dh_\alpha}{dt} = c_{p\alpha} \frac{dT}{dt}$, yielding

$$\begin{aligned}
\frac{d}{dt}(\Sigma_\alpha \rho_\alpha u_\alpha) &= \Sigma_\alpha \left(\rho_\alpha c_{p\alpha} \frac{dT}{dt} + \frac{dp_\alpha}{dt} \left(\frac{p_\alpha}{\rho_\alpha} + u_\alpha \right) - \frac{dp_\alpha}{dt} \right) \\
&= \underbrace{(\rho c_p)_{\text{eff}} \frac{dT}{dt}}_{\text{energy storage due to temperature change}} + \underbrace{\Sigma_\alpha \left(\frac{dp_\alpha}{dt} \left(\frac{p_\alpha}{\rho_\alpha} + u_\alpha \right) - \frac{dp_\alpha}{dt} \right)}_{\text{energy storage due to pressure work or density change, e.g. by phase transition}}.
\end{aligned} \quad (33)$$

[†] $\frac{d\rho_{LR}}{dt} = \frac{\partial \rho_{LR}}{\partial p_{LR}} \frac{dp_{LR}}{dt} + \frac{\partial \rho_{LR}}{\partial T} \frac{dT}{dt}$ and $\frac{d\rho_{GR}^W}{dt} = \frac{\partial \rho_{GR}^W}{\partial p_{GR}} \frac{dp_{GR}}{dt} + \frac{\partial \rho_{GR}^W}{\partial T} \frac{dT}{dt}$.

Table 1

Explanation of numerical models and their abbreviations used in this work, cf. Fig. 1.

Model abbreviation	Description
TH ² M	The model featuring non-isothermal two-phase (liquid and gaseous phase) two-component (water and non-aqueous component) flow with linear poro-elasticity. ²⁰ For the model application in this work, phase transition of the water component is allowed, but dissolution of the non-aqueous component into the liquid phase is prevented. This model represents the highest physical complexity considered in this study, but the implementation allows for even higher complexity if the dissolution of the non-aqueous component is allowed.
TH ² M-R	The “Richards-TH ² M model”, wherein the gas pressure is constrained to atmospheric levels on the entire domain throughout the simulation. The water component can exist in both phases, whereas the non-aqueous component is restricted from dissolving into the liquid phase. This model represents a simplification to the higher complexity of the full TH ² M model.
TH ² M-S	The “single phase TH ² M model”. Full liquid saturation is enforced, and non-aqueous component dissolution continues to be prevented. This model represents the most simplified TH ² M model in this study.
TRM	The non-isothermal Richards mechanics implementation based on the non-isothermal Richards flow equation presented in Ref. 19 coupled to linear poro-elasticity. It includes a vapour diffusion model and can be used to model both saturated and unsaturated problems.
TRM-S	The “single phase TRM model” wherein full liquid saturation is enforced. Therefore, there exists no gaseous phase and no water vapour. This model represents a simplification to the TRM model.

Table 2Comparison of the water (component) mass balance equations as implemented in the TH²M and TRM models.

Term/process	TH ² M implementation	TRM implementation
Deformation	$0 = ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})\alpha_B \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$	$0 = S_L\rho_{LR}\alpha_B \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$
Advection	$+ \operatorname{div} (\rho_{LR}\tilde{\mathbf{w}}_{LS} + \rho_{GR}^W\tilde{\mathbf{w}}_{GS})$	$+ \operatorname{div} (\rho_{LR}\tilde{\mathbf{w}}_{LS})$
Diffusion	$+ \operatorname{div} \left[-\rho_{GR}^W \mathbf{D}_G^W \left(\frac{\partial x_{m,G}^W}{\partial p_{GR}} \operatorname{grad} p_{GR} + \frac{\partial x_{m,G}^W}{\partial p_{cap}} \operatorname{grad} p_{cap} + \frac{\partial x_{m,G}^W}{\partial T} \operatorname{grad} T \right) \right]$	$+ \operatorname{div} \left(-\mathbf{D}_G^W \left(\frac{\partial p_{GR}^W}{\partial p_{LR}} \operatorname{grad} p_{LR} + \frac{\partial p_{GR}^W}{\partial T} \operatorname{grad} T \right) \right)$
Thermal expansion	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \phi S_L \rho_{LR} \beta_{T,LR} \right\} \frac{dT}{dt}$ $- ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) (\alpha_B - \phi) \operatorname{tr} (\boldsymbol{\alpha}_{T,SR}) \left\} \frac{dT}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \rho_{LR} S_L \phi \beta_{T,LR} \right\} \frac{dT}{dt}$ $- \rho_{LR} S_L (\alpha_B - \phi) \operatorname{tr} (\boldsymbol{\alpha}_{T,SR}) \left\} \frac{dT}{dt}$
Storage/saturation	$+ [\phi(\rho_{LR} - \rho_{GR}^W) - ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) p_{cap} (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}$	$+ [\phi(\rho_{LR} - \rho_{GR}^W) + (\rho_{LR} - \rho_{GR}^W) \rho_{LR} S_L (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}$
Storage/compression due to $\frac{dp_{cap}}{dt} / \frac{dp_{LR}}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{cap}} - \phi S_L \rho_{LR} \beta_{p,LR} \right\} \frac{dp_{cap}}{dt}$ $- ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} S_L \left\} \frac{dp_{cap}}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{LR}} + \rho_{LR} S_L \phi \beta_{p,LR} \right\} \frac{dp_{LR}}{dt}$ $+ \rho_{LR} S_L (\alpha_B - \phi) \beta_{p,SR} S_L \left\} \frac{dp_{LR}}{dt}$
Storage/compression due to $\frac{dp_{GR}}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{GR}} + \phi S_L \rho_{LR} \beta_{p,LR} \right\} \frac{dp_{GR}}{dt}$ $+ ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} \left\} \frac{dp_{GR}}{dt}$	

Table 3Comparison of energy balance equations as implemented in the TH²M and TRM models.

Term/process	TH ² M implementation	TRM implementation
Storage	$0 = \frac{d}{dt} (\Sigma_\alpha \rho_\alpha u_\alpha)$	$0 = (\rho c_p)_{eff} \frac{dT}{dt} + L_0 \frac{d\theta_{vap}}{dt}$
Heat conduction	$- \operatorname{div} (\lambda_{eff} \operatorname{grad} T)$	$- \operatorname{div} (\lambda_{eff} \operatorname{grad} T)$
Energy advected by phase flux	$+ \operatorname{div} (h_L \mathbf{A}_L + h_G \mathbf{A}_G)$	$+ \operatorname{grad} T \cdot c_{pL} \mathbf{A}_L$
Energy advected by diffusive flux	$+ \operatorname{div} (h_G^W \mathbf{J}_G^W + h_G^C \mathbf{J}_G^C)$	$+ c_{p,vap} \mathbf{J}_G^W \cdot \operatorname{grad} T$ $+ \operatorname{div} (L_0 \mathbf{J}_G^W / \rho_{LR})$
Work of gravitation	$- \Sigma_\alpha \mathbf{g} \cdot \mathbf{A}_\alpha$	
Medium volume change	$+ (\Sigma_\alpha \rho_\alpha h_\alpha) \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$	

Thus, it is clear that the energy storage due to temperature changes is equivalent in the two implementations, while the latent heat related to the vaporisation of water is expressed by pressure and density derivatives in TH²M by the last terms of the above equation. The formulations in the two models are quite different. However, some equivalence can be found in most of the terms and the numerical results will demonstrate that in the discussed examples, any differences arising from the different formulations are small enough to be neglected in the discussion of results.

Finally, for the comparison of the momentum balance equations, Eq. (23) can be used. The two equations represent the same process but in TH²M, the gas pressure and gaseous phase density is considered. This has a high impact for high-gas-pressure systems, e.g. in gas injection experiments. In systems of lower gas pressures, the impact is expected to be much lower and will be discussed in the following Section 4.

3.2. Richards case: Constant p_{GR} with vapour diffusion

In this case, the complexity of the full TH²M model is reduced to the TH²M-R model (Table 1), while the phase transition of the water component between the liquid and the gaseous phase remains allowed and the dissolution of the non-aqueous component into the liquid phase remains disregarded. To match assumptions intrinsic to the TRM model in the TH²M-R model, the gas pressure p_{GR} is assumed to be spatially and temporally constant. The following assumptions can thus be made for the rearranged TH²M water component mass balance Eq. (31):

- no advective component flux in gaseous phase $\mathbf{A}^W = \mathbf{A}_L^W + \mathbf{A}_G^W = \mathbf{A}_L^W = \rho_{LR} \tilde{\mathbf{w}}_{LS} = \rho_{LR} \tilde{\mathbf{w}}_{LS}$,
- constant p_{GR} , therefore $\frac{dp_{GR}}{dt} = 0$ and $\operatorname{grad} p_{GR} = \mathbf{0}$.

As a consequence, the density derivatives from Eq. (40) simplify with regard to their gas pressure derivatives, and the diffusive flux of the water component in the gaseous phase Eq. (7) simplifies to:

$$\mathbf{J}^\zeta = -\rho_{GR}^W \mathbf{D}_G^\zeta \left(\frac{\partial x_{m,G}^W}{\partial p_{cap}} \operatorname{grad} p_{cap} + \frac{\partial x_{m,G}^W}{\partial T} \operatorname{grad} T \right). \quad (34)$$

Table 4Comparison of the water mass balance equations with Richards assumptions implicit in TRM and explicit in TH²M-R.

Term/process	TH ² M-R	TRM
Deformation	$0 = ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})\alpha_B \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$	$0 = S_L\rho_{LR}\alpha_B \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$
Advection	$+ \operatorname{div} (\rho_{LR}\tilde{\mathbf{u}}_{LS})$	$+ \operatorname{div} (\rho_{LR}\tilde{\mathbf{u}}_{LS})$
Diffusion	$+ \operatorname{div} \left[-\rho_{GR}D_G^W \left(\frac{\partial x_{m,G}^W}{\partial p_{cap}} \operatorname{grad} p_{cap} + \frac{\partial x_{m,G}^W}{\partial T} \operatorname{grad} T \right) \right]$	$+ \operatorname{div} \left(-D_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{LR}} \operatorname{grad} p_{LR} + \frac{\partial \rho_{GR}^W}{\partial T} \operatorname{grad} T \right) \right)$
Thermal expansion	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \phi S_L \rho_{LR} \beta_{T,LR} - ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})(\alpha_B - \phi) \operatorname{tr}(\alpha_{T,SR}) \right\} \frac{dT}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \rho_{LR} S_L \phi \beta_{T,LR} - \rho_{LR} S_L (\alpha_B - \phi) \operatorname{tr}(\alpha_{T,SR}) \right\} \frac{dT}{dt}$
Storage/saturation	$+ [\phi(\rho_{LR} - \rho_{GR}^W) - ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) p_{cap}(\alpha_B - \phi)\beta_{p,SR}] \frac{dS_L}{dt}$	$+ [\phi(\rho_{LR} - \rho_{GR}^W) + (\rho_{LR} - \rho_{GR}^W) \rho_{LR} S_L (\alpha_B - \phi)\beta_{p,SR}] \frac{dS_L}{dt}$
Storage/compressibilities	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{cap}} - \phi S_L \rho_{LR} \beta_{p,LR} - ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})(\alpha_B - \phi)\beta_{p,SR} S_L \right\} \frac{dp_{cap}}{dt}$	$+ \left\{ \phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{LR}} + \rho_{LR} S_L \phi \beta_{p,LR} + \rho_{LR} S_L (\alpha_B - \phi)\beta_{p,SR} S_L \right\} \frac{dp_{LR}}{dt}$

If we apply these assumptions to the component mass balance Eq. (31) or Eq. (2), we get:

$$\begin{aligned}
0 = & \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{cap}} - \phi S_L \rho_{LR} \beta_{p,LR} \right. \\
& - \left. ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})(\alpha_B - \phi)\beta_{p,SR} S_L \right) \frac{dp_{cap}}{dt} \\
& + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \phi S_L \rho_{LR} \beta_{T,LR} \right. \\
& - \left. ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})(\alpha_B - \phi) \operatorname{tr}(\alpha_{T,SR}) \right) \frac{dT}{dt} \\
& + ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR})\alpha_B \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \operatorname{div} (\rho_{LR}\tilde{\mathbf{u}}_{LS}) \\
& + \operatorname{div} \left[-\rho_{GR}D_G^W \left(\frac{\partial x_{m,G}^W}{\partial p_{cap}} \operatorname{grad} p_{cap} + \frac{\partial x_{m,G}^W}{\partial T} \operatorname{grad} T \right) \right] \\
& + [\phi(\rho_{LR} - \rho_{GR}^W) - ((1 - S_L)\rho_{GR}^W + S_L\rho_{LR}) p_{cap}(\alpha_B - \phi)\beta_{p,SR}] \frac{dS_L}{dt}.
\end{aligned} \quad (35)$$

This TH²M-R water component mass balance with the Richards assumptions explicitly taken into account (Eq. (35)) is compared with the TRM water mass balance equation (Eq. (32)) in Table 4 and discussed below. Regarding the momentum balance Eqs. (23), we stated in Section 2.2 that differences between TRM and TH²M-R arise only from the gas pressure p_{GR} and gas density ρ_{GR} . If the gaseous phase density for $p_{GR} = 1$ atm is estimated to be around three orders of magnitude smaller than liquid phase density, its contribution to the gravity-induced stress divergence is expected to be very small. However, the contribution of a non-zero gas pressure to the equilibrium conditions can only be neglected for full saturation and conditions where $|\rho_{GR}| \ll |\rho_{LR}|$. The term $\operatorname{div} [\alpha_B(1 - \chi)p_{GR}\mathbf{I}]$ becomes non-zero in the presence of saturation gradients so that a momentum transfer to the gas phase occurs even for constant gas pressure. It may be practical to transition to a net-stress formulation under these conditions:

$$\mathbf{0} = \begin{cases} \text{TH}^2\text{M-R :} & \operatorname{div} (\boldsymbol{\sigma}_{\text{net}} + \alpha_B \chi p_{cap} \mathbf{I}) \\ & + [(1 - \phi)\rho_{SR} + S_L \phi \rho_{LR} + (1 - S_L) \phi \rho_{GR}] \mathbf{b} \\ \text{TRM :} & \operatorname{div} (\boldsymbol{\sigma}_{\text{net}} - \alpha_B \chi p_{LR} \mathbf{I}) + [(1 - \phi)\rho_{SR} + S_L \phi \rho_{LR}] \mathbf{b}, \end{cases} \quad (36)$$

where $\boldsymbol{\sigma}_{\text{net}} = \boldsymbol{\sigma}' - p_{GR}\mathbf{I}$. Therefore, the momentum balance equations in the TH²M-R and TRM implementations can be considered equivalent – with the exception of the gas density contribution to the gravity-induced stresses.

Table 5Comparison of energy balance equations for the TH²M-R and TRM models.

Term/process	TH ² M-R	TRM
Storage	$0 = \frac{d}{dt} (\sum_a \rho_a u_a)$	$0 = (\rho c_p)_{\text{eff}} \frac{dT}{dt} + L_0 \frac{d\theta_{\text{vap}}}{dt}$
Heat conduction	$- \operatorname{div} (\boldsymbol{\lambda}_{\text{eff}} \operatorname{grad} T)$	$- \operatorname{div} (\boldsymbol{\lambda}_{\text{eff}} \operatorname{grad} T)$
Energy advected by phase flux	$+ \operatorname{div} (h_L \mathbf{A}_L)$	$+ c_{p,L} \mathbf{A}_L \operatorname{grad} T$
Energy advected by diffusive flux	$+ \operatorname{div} (h_G^W \mathbf{J}_G^W + h_G^C \mathbf{J}_G^C)$	$+ c_{p,\text{vap}} \mathbf{J}_G^W \cdot \operatorname{grad} T + \operatorname{div} (L_0 \mathbf{J}_G^W / \rho_{LR})$
Work of gravitation	$- \mathbf{g} \cdot \mathbf{A}_L$	
Medium volume change	$+ (\sum_a \rho_a h_a) \operatorname{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$	

Regarding the energy balance equation, one can see that the TH²M-R advective energy transport simplifies, when the Richards assumptions are considered: $\mathbf{A}_G = 0$. Thus, the comparison of the TH²M-R and TRM energy balances is shown in Table 5, where we can observe that energy transport by component diffusion or advection are expressed similarly in both implementations. The energy storage terms both take into account the effective specific heat capacity. While water vaporisation in TRM is expressed by the empirical relation Eq. (29), in TH²M and TH²M-R it is expressed via liquid density and pressure changes.

The water component mass balance equation of TH²M-R is nearly identical to the water mass balance of the TRM implementation. The two PDEs are compared in Table 4, and we note the following observations regarding each term:

- Deformation. The two models approach each other for low vapour densities and high liquid saturation. As opposed to the “general case” with free gas pressure, there is no gas-pressure-induced deformation in the “Richards case”.
- Advection. The formulations of the two models are identical.
- Diffusion. In both models, the diffusion is driven by temperature gradients and gradients of p_{cap} (TH²M-R) or p_{LR} (TRM). With constant p_{GR} , it follows from Eq. (1) that $\operatorname{grad} p_{cap} = -\operatorname{grad} p_{LR}$, so the diffusion models can be considered equivalent in this regard. However, since vapour diffusion is expressed in terms of mass fractions $x_{m,G}^W$ in the TH²M-R model and vapour densities in the TRM model, some differences arise. The TH²M-R formulation can be expressed in terms of vapour pressure (see Appendix B for details), yielding:

$$\mathbf{J}_G^W = -\frac{M_C}{M_G} D_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{cap}} \operatorname{grad} p_{cap} + \left(\frac{\rho_{GR}^W}{T} + \frac{\partial \rho_{GR}^W}{\partial T} \right) \operatorname{grad} T \right). \quad (37)$$

It can be seen that the factor in front of $\text{grad } T$ in Eq. (37) is always greater than the equivalent factor in the TRM vapour diffusion. Since in the numerical examples in this work, we consider water ($M_W = 18 \times 10^{-3} \text{ kg mol}^{-1}$) and air ($M_C = 28 \times 10^{-3} \text{ kg mol}^{-1}$) as the two components and $M_G = \sum_{\zeta} x_{n,G}^{\zeta} M_{\zeta}$, the factor $M_C M_G^{-1}$ is always greater than one. For these two reasons, one can expect the diffusive vapour flux predicted by the TH²M-R model to be greater than its equivalent in TRM.

- The observations made in the “general case” (cf. Table 2) regarding thermal expansion, storage due to partial saturation and storage due to compression hold true for the “Richards case” with TH²M-R and TRM as well. The expressions in the two models approach each other for low vapour densities and high liquid saturation.

3.3. Single phase fully saturated case

In this case, full saturation is assumed, and we find that this case corresponds to a single (liquid) phase region. The TH²M (or TH²M-R) model reduces in complexity to the TH²M-S model, and the TRM model reduces to the TRM-S model (Table 1). Without dissolved gas (or vaporised water, respectively), the following modifications can be made to the balance equation describing the constituent water, $\zeta \equiv W$:

- there is no gas phase present; related density measures vanish,
- full saturation: $S_L = 1$ and $\frac{dS_L}{dt} = 0$,
- only the water component is present, so component designators are dropped: $\rho_{FR}^{\zeta} = \rho_{LR}^{\zeta} = \rho_{LR}$,
- no diffusive-dispersive flux because only a single component is present: $\mathbf{J}^{\zeta} = 0$,
- mass transport occurs by (advective) liquid phase mass flux: $\mathbf{A}^{\zeta} = \rho_{LR} \tilde{\mathbf{w}}_{LS}$.

These assumptions can be inserted into Eq. (35) (or into Eqs. (31) or (2) for that matter) for the TH²M-S model and into Eq. (19) for the TRM-S model, yielding:

$$0 = -\rho_{LR} (\phi \beta_{T,LR} + (\alpha_B - \phi) \text{tr}(\alpha_{T,SR})) \frac{dT}{dt} + \rho_{LR} \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div}(\rho_{LR} \tilde{\mathbf{w}}_{LS}) \begin{cases} \text{TH}^2\text{M} : -\rho_{LR} (\phi \beta_{p,LR} + (\alpha_B - \phi) \beta_{p,SR}) \frac{dp_{\text{cap}}}{dt} \\ \text{TRM} : +\rho_{LR} (\phi \beta_{p,LR} + (\alpha_B - \phi) \beta_{p,SR}) \frac{dp_{LR}}{dt} \end{cases} \quad (38)$$

Regarding the energy balance equation in TH²M-S, all the assumptions from Section 3.2 hold true. Additionally, enthalpy measures related to components in the gaseous phase vanish, and in the TRM implementation Eq. (28), we insert $\mathbf{J}_G^W = 0$ and $\frac{d\theta_{\text{vap}}}{dt} = 0$. Thus, the energy balance equations collapse to the formulation given in Table 6. Regarding the momentum balance equations Eq. (23), $S_L = 1$ and $\chi = 1$ can be inserted into the TH²M momentum balance, making all gaseous phase related variables vanish and rendering the TH²M-S momentum balance identical to the TRM-S momentum balance. The TRM momentum balance remains unaffected by the full saturation assumption and therefore, the TRM-S and TRM momentum balance equations are identical.

4. Numerical examples

In order to illustrate and discuss the comparison of the various model formulations, the three cases discussed in Section 3 are simulated. In the theoretical section above, assumptions were added gradually, increasingly simplifying the governing equations from one case to the next. In the following, the complexity evolves in the opposite direction: We start with the simplest case of full saturation in a homogeneous domain and move on to unsaturated and heterogeneous problems. In the DECOVALEX 2023 project,¹⁷ Task C deals with the numerical THM modelling of the FE experiment conducted at the Mont Terri URL

Table 6

Comparison of energy balance equations for the TH²M-S and TRM-S models.

Term/process	TH ² M-S	TRM-S
Storage	$0 = \frac{d}{dt} (\sum_a \rho_a u_a)$	$0 = (\rho c_p)_{\text{eff}} \frac{dT}{dt}$
Heat conduction	$-\text{div}(\lambda_{\text{eff}} \text{grad } T)$	$-\text{div}(\lambda_{\text{eff}} \text{grad } T)$
Energy advected by phase flux	$+\text{div}(h_L \mathbf{A}_L)$	$+c_{pL} \mathbf{A}_L \text{grad } T$
Work of gravitation	$-\mathbf{g} \cdot \mathbf{A}_L$	
Medium volume change	$+\rho_L h_L \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right)$	

Table 7

Measures of the 2D geometry^{33,44}.

Description	Symbol in Fig. 2	Value	Reference
Radius of FE tunnel	r_{tunnel}	2.48 m	Nagra, 2019
Radius of heater	r_{heater}	1.05 m	Nagra, 2019
Opalinus Clay domain	—	50 m × 50 m	—

in Switzerland.^{33,44} The first steps of the modelling work comprise coupled 2D THM simulations of the heating phase, demonstrating the initial increase of saturation of the bentonite backfill, its subsequent dry-out in the near-field of the heater due to the heating and eventually resaturating again due to liquid flow from the fully saturated host rock (Opalinus Clay). The examples used in the subsequent sections are motivated by these analyses.

4.1. Description of numerical examples

4.1.1. Geometry

A detailed description of the examples is given in Ref. 44. Here, an overview and differences to what is specified there are given. The domain consists of a 50 m × 50 m square with a circle in the centre, representing the tunnel. In the middle of the tunnel, there is a circular hole, representing the heater. There are two materials present: In the tunnel, there is granular bentonite acting as backfill, and surrounding it, there is the host rock Opalinus Clay (Fig. 2). The area (or volume) of the heater itself is not modelled. Instead, it is represented by boundary conditions imposed at the inner boundary of the tunnel. Thus, a possible deformation of the heater and its heat capacity are neglected in the examples. The dimensions of heater, tunnel and domain are given in Table 7.

4.1.2. Boundary conditions and initial conditions

No hydraulic flow can occur across the inner and outer boundaries representing the heater and the domain boundaries. Likewise, no heat flow across the outer boundaries is allowed. Mechanical boundaries comprise zero displacement in the x - and y -direction at any outer or inner boundary. To represent the heating, a Neumann boundary condition is imposed on the inner boundary with a heating power of 89 W m^{-2} . In the TH²M model, a gas pressure is initialised at atmospheric values, $p_{GR}(t = 0) = 1.013 \times 10^5 \text{ Pa}$. For the TH²M-R model in the “Richards case” with constant gas pressure, a boundary condition is imposed on the entire domain, holding the gas pressure constant at the initial value. The initial temperature of the entire domain is 15°C . The initial total stresses cause no initial displacements and are 2.5 MPa in the x -direction and 6.5 MPa in the y -direction. The initial liquid pressure in the Opalinus Clay is defined as 2.0 MPa for the TRM model. For the TH²M models, this means $p_{\text{cap}} = p_{GR} - p_{LR} = -1.8987 \text{ MPa}$ as initial capillary pressure. The initial saturation in the granular bentonite is 23%, leading to an initial capillary pressure of 123 MPa.

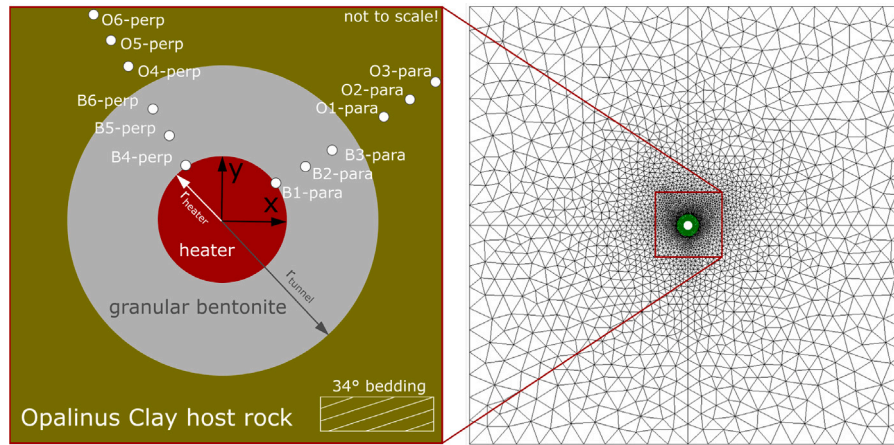


Fig. 2. Geometry of the domain and finite element mesh. The domain extends to $50\text{ m} \times 50\text{ m}$ and contains two materials: Opalinus Clay in the outer area and granular bentonite in the tunnel around the heater. The heater itself is not modelled, but rather represented by inner boundary conditions.

Table 8
Material properties for the three case studies (modified after⁴⁴).

Input parameters	Symbol	Unit	OPA	GBM
Thermal conductivity, parallel to bedding, dry	$\lambda_{\text{dry},\parallel}$	$\text{W m}^{-1} \text{K}^{-1}$	2.4	0.35
Thermal conductivity, perpendicular to bedding, dry	$\lambda_{\text{dry},\perp}$	$\text{W m}^{-1} \text{K}^{-1}$	1.3	0.35
Thermal conductivity, parallel to bedding, saturated	$\lambda_{\text{sat},\parallel}$	$\text{W m}^{-1} \text{K}^{-1}$	2.4	1.2
Th. conductivity, perpendicular to bedding, saturated	$\lambda_{\text{sat},\perp}$	$\text{W m}^{-1} \text{K}^{-1}$	1.3	1.2
Solid specific heat capacity	c_p	$\text{J kg}^{-1} \text{K}^{-1}$	995.0	800.0
Solid density	ρ_{SR}	kg m^{-3}	2689.66	2227.2
Porosity	ϕ	–	0.13	0.331
Permeability, parallel to bedding	$k_{\text{S}\parallel}$	m^2	5.0×10^{-20}	3.5×10^{-20}
Permeability, perpendicular to bedding	$k_{\text{S}\perp}$	m^2	1.0×10^{-20}	3.5×10^{-20}
Van Genuchten gas entry pressure	p_c	MPa	20.0	28.6
Van Genuchten exponent	n	–	2.5	2.0
Residual liquid saturation	$S_{\text{L, res}}$	–	0.0	0.0
Maximum liquid saturation	$S_{\text{L, max}}$	–	0.9999	0.9999
Relative liquid phase permeability	$k_{\text{L}}^{\text{rel}}$	–	Eq. (18)	
Relative gaseous phase permeability	$k_{\text{G}}^{\text{rel}}$	–	1.0	1000.0
Young's modulus, parallel to bedding	$E_{\parallel}$	MPa	8000	18.0
Young's modulus, perpendicular to bedding	$E_{\perp}$	MPa	4000	18.0
Shear modulus, perpendicular to bedding	$G_{\perp}$	MPa	3500	6.667
Poisson's ratio, parallel to bedding	$\nu_{\parallel}$	–	0.35	0.35
Poisson's ratio, perpendicular to bedding	$\nu_{\perp}$	–	0.25	0.35
Linear solid thermal expansion (isotropic)	$\alpha_{\text{T, SR}}$	K^{-1}	1.5×10^{-5}	3.0×10^{-6}
Biot–Willis coefficient	α_{B}	–	1.0	
Bishop's coefficient	χ	–	$\chi = S_{\text{L}}$	
Reference liquid phase density	$\rho_{\text{LR, ref}}$	kg m^{-3}	1000.0	
Reference liquid phase pressure	$p_{\text{LR, ref}}$	MPa	2.0	
Reference temperature	T_{ref}	K	288.15	
Water component specific heat capacity	c_p^{W}	$\text{J kg}^{-1} \text{K}^{-1}$	4181.3	
Non-aqueous component specific heat capacity	c_p^{G}	$\text{J kg}^{-1} \text{K}^{-1}$	733.0	
Liquid phase viscosity	μ_{LR}	Pa s	Eq. (17)	
Gaseous phase viscosity	μ_{GR}	Pa s	1.0×10^{-5}	
Water component molar mass	M_{W}	kg mol^{-1}	0.018	
Non-aqueous component molar mass	M_{G}	kg mol^{-1}	0.0285	
Liquid phase linear thermal expansion coefficient	$\beta_{\text{T, LR}}$	K^{-1}	4.0×10^{-4}	
Water vapour diffusivity in gaseous phase	D_{G}^{W}	$\text{m}^2 \text{s}^{-1}$	2.15×10^{-5}	

4.1.3. Parameterisation

A list of thermal, mechanical and hydraulic material properties as well as a definition of bulk and fluid properties is given in Table 8. The bentonite has isotropic properties, whereas the Opalinus Clay is transversely isotropic with respect to its bedding plane dipping towards the left with an inclination of 34° against the horizontal. Both materials are modelled as linear elastic for simplicity.

4.1.4. Evaluation points and numerical settings

With the given geometry (Fig. 2 and Table 7), Gmsh⁴⁵ was used to create the finite element mesh. The finite element type was a triangular

Taylor-Hood element type (quadratic order elements for the mechanical process and linear order elements for the hydraulic and thermal processes). The resulting non-linear monolithic equations were numerically solved using a Newton–Raphson linearisation with a convergence criterion on the L^2 -norm of the increment of each of the primary unknowns set to relative tolerances of 1×10^{-6} . The linear system of equations was then solved by an implementation of the BiCGSTAB solver, a stabilised version of the bi-conjugate gradient method, with ILUT preconditioning.⁴⁶ The temporal discretisation was done by the backward Euler method with time step size being a function of the time: The simulation starts at $t = 0\text{ s}$ with 25 fixed time steps of $1 \times 10^{-2}\text{ s}$, followed by 25 fixed time steps of $1 \times 10^{-1}\text{ s}$, followed by 25 fixed time

Table 9
List of evaluation points (modified after⁴⁴).

Point	x-coordinate	y-coordinate	Description
B1-para	0.44 m	0.30 m	Heater surface
B2-para	0.60 m	0.41 m	Tunnel near-field
B3-para	0.86 m	0.58 m	Tunnel far-field
O1-para	4.15 m	2.80 m	Opalinus Clay near-field
O2-para	6.63 m	4.47 m	Opalinus Clay mid-field
O3-para	11.61 m	7.83 m	Opalinus Clay far-field
B4-perp	-0.29 m	0.44 m	Heater surface
B5-perp	-0.41 m	0.60 m	Tunnel near-field
B6-perp	-0.58 m	0.86 m	Tunnel far-field
O4-perp	-2.80 m	4.15 m	Opalinus Clay near-field
O5-perp	-4.47 m	6.63 m	Opalinus Clay mid-field
O6-perp	-7.83 m	11.61 m	Opalinus Clay far-field

steps of 1 s and so on until the simulation is stopped at $t = 1.6 \times 10^8$ s. To evaluate and compare simulation results, the observation points defined in Table 9 were chosen.

4.2. Fully saturated single phase case – homogeneous properties

In this case, all materials are assigned Opalinus Clay properties and initial conditions, i.e. fully saturated conditions throughout the entire simulation. The TRM-S and TH²M-S results of this test case are shown in Figs. 3 and 4. The THM-coupled process is driven by the heat production of the central boundary representing the heater. Initially, heat transport occurs by conduction only. The increasing temperature of the liquid and grain skeleton causes a thermal pressurisation of the liquid due to differential thermal expansion of the liquid phase and the solid (grains and skeleton).²⁸ The rock expansion due to a local temperature increase simultaneously induces a passive extension farther away, causing a pressure decrease prior to the expected increase upon arrival of the temperature front at a later time (e.g. liquid pressure results at point O1-para in Fig. 4). In the far-field, the expansion of the near-field leads to a compression of the medium, since the domain is confined by the outer zero-displacement boundary conditions. As temperature continues to increase, the liquid drains along the pressure gradient. The results obtained by the TRM-S and TH²M-S models are in agreement because as shown in Section 3.3, the water component balance equations reduce to a THM equation which is identical for both implementations. The same holds true for the momentum balance equations.

4.3. Richards case: Constant p_{GR} with vapour diffusion

In this case, the materials and initial conditions are simulated as defined in the model description above and the results of the TRM and TH²M-R models are shown in Figs. 5 and 6. In the Opalinus Clay, the initial liquid pressure is 2 MPa, whereas in the granular bentonite in the FE-tunnel, the unsaturated initial condition leads to an initial capillary pressure of about 123 MPa (see Section 4.1.2). In the bentonite, this setting leads to two opposing processes: On the one hand, the strong liquid pressure gradient at the interface of the bentonite and the host rock is the driving force of liquid flow from the host rock towards the centre of the tunnel. On the other hand, the temperature gradient created by the heat source drives the vapour diffusion and thus water component transport away from the centre of the tunnel. These two opposing mechanisms lead to a complex interplay of dry-out and resaturation in the near-field of the tunnel (heat pipe effect). As demonstrated in Section 3.2, the water component mass balances reduce to a nearly-identical formulation if the Richards assumptions are applied to the TH²M water component mass balance. Differences may arise only from the vapour diffusion term which is expressed using the vapour density in TRM whereas in the TH²M (TH²M-R) implementation, the mass fraction is used as discussed in Section 3.2,

Eq. (37). In Eqs. (46) and (37), the vapour diffusion coefficient D_G^W is scaled by a factor $M_C M_G^{-1}$, which is close to one if there is no water vapour in the gaseous phase. In this study, the non-aqueous component was assigned a molar mass equivalent to a normal air mixture, which is higher than the water molar mass. As a consequence, $M_C M_G^{-1}$ will be greater than one if there is water vapour present because the average molar mass of the gas mixture $M_G = \sum_{\zeta} x_{\zeta, G} M_{\zeta}$ decreases as a function of increasing water vapour content. Additionally, there is one extra term present in the thermal diffusion part of the TH²M-R model, which is also always greater than zero. Therefore, it is expected that when using identical diffusion coefficients in the TH²M-R and TRM models that the diffusion in the former is slightly stronger.

Fig. 5 shows the evolution of some of the main variables at the observation points in the bentonite. While temperature and displacement results obtained by the two models are nearly identical, some slight differences are observable with respect to the liquid pressure evolution. The points farthest from the heater and closest to the host rock show nearly identical pressure evolution and illustrate how the saturation front moves from the Opalinus Clay into the bentonite. The points closer to the heater show an initial decrease in liquid pressure corresponding to the dry out before subsequently resaturating. As expected, the TH²M-R model predicts slightly stronger desaturation as the driving force of the dry-out – the thermal diffusion – is slightly stronger and diffusion increases with the decreasing molar mass of the gaseous phase.

Fig. 6 illustrates the evolution of the main variables in the far-field, where the match of the two model predictions is very good. This is to be expected since the host rock is initially fully saturated and thus the water mass balance simplifies as illustrated in the previous section. When comparing these results to the fully saturated “single phase” case (Fig. 4), it can be seen that liquid pressures are generally lower because liquid is able to flow from the host rock into the bentonite. Figs. 5 and 6 also show the gas pressure result of the TH²M-R model, which is kept constant throughout the simulation for an explicit modelling of the Richards assumption.

4.4. General case: Full TH²M and TH²M-R models

In this last test case, there are no additional simplifying assumptions made for the TH²M model and the gas pressure is allowed to evolve freely. For completeness, the comparison of results obtained by the TRM and TH²M models is given in Appendix D, Figs. 10 and 11. However, in the following, we will compare TH²M-R results obtained with constant fixed gas pressures against results obtained by TH²M with free gas pressures. This is done in order to eliminate any differences of the results that may originate from slightly different implementations (see section above). Any difference in the results can thus be attributed to the freely evolving gas pressure. Fig. 8 shows and compares the results from the outer parts of the domain in the Opalinus Clay obtained by the TH²M and TH²M-R models. Firstly, it can be stated that the results are nearly identical, with the obvious exception of the gas pressure evolution itself. Very slight differences are visible in the liquid pressures at later stages of the simulation, which are more pronounced than in the fully saturated and Richards cases discussed before, e.g. with regard to the displacements in Fig. 8. With a free gas pressure, the liquid pressures are slightly lower and – likely as a consequence of the HM-coupling – the displacements of respective observation points show slightly lower amplitudes as well. The reasons for this can be observed when looking at the near-field of the heater in Fig. 7. There, it can be seen that the dry-out in the TH²M-R results is more prominent than in the TH²M results. Consequently, more water is transported away from the heater towards the outer border of the bentonite and into the Opalinus Clay, which in turn increases the saturation resp. liquid pressure there a bit more than in the TH²M results. The resulting liquid pressure gradient between bentonite and host rock becomes therefore larger in TH²M-R, and the liquid flows into the host rock a little bit

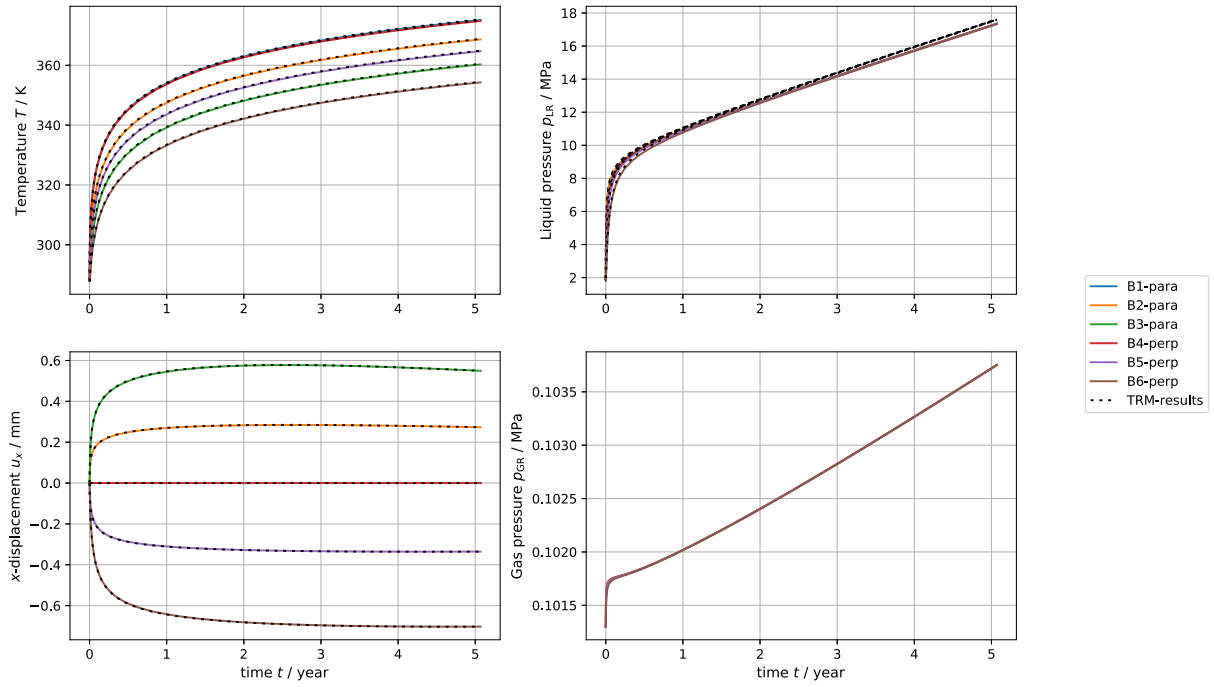


Fig. 3. Comparison of results obtained in the tunnel by the TRM-S (black dashed) and TH²M-S (coloured solid) models in the fully saturated, homogeneous case with Opalinus Clay properties everywhere. The TH²M-S model obtains non-zero gas pressures in the residual gas phase. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

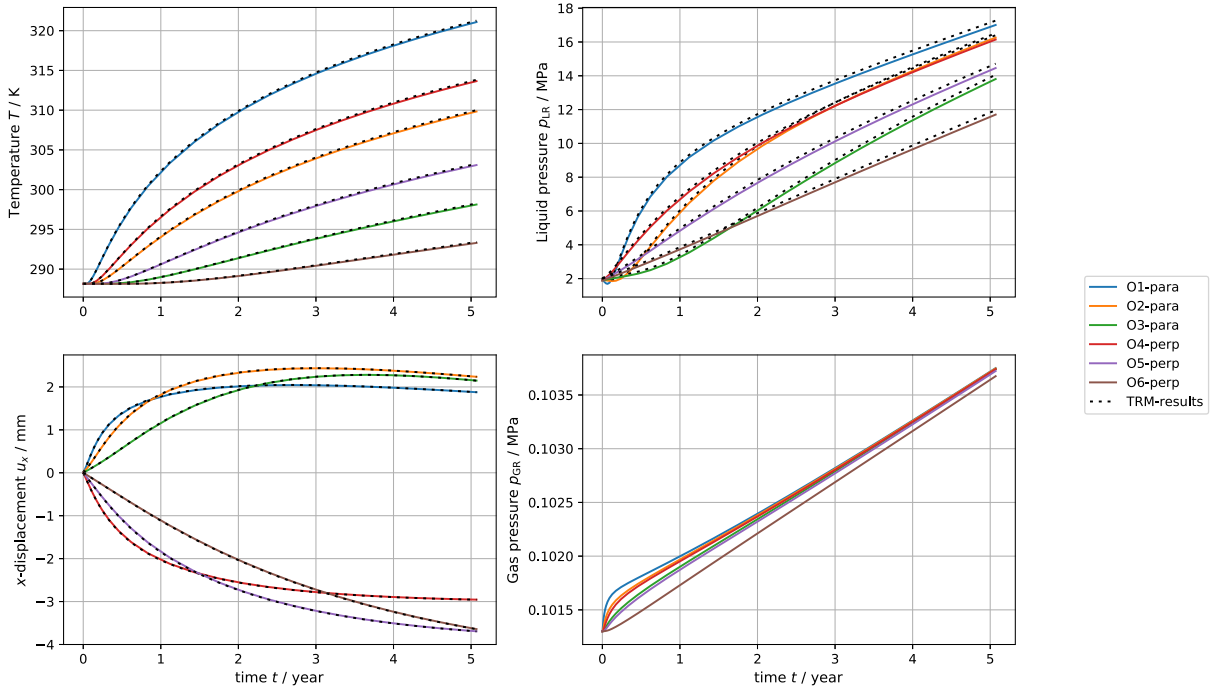


Fig. 4. Comparison of results obtained in the Opalinus Clay by the TRM-S (black dashed) and TH²M-S (coloured solid) models in the fully saturated, homogeneous case with Opalinus Clay properties everywhere. The TH²M-S model obtains non-zero gas pressures in the residual gas phase. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

slower in the TH²M results. In short, a stronger vapour diffusion like in the TH²M-R results pushes more water away from the heater (see also the previous section).

The reason for the weaker diffusive transport of the water component in the TH²M model can be found in the definition of the mass fraction $x_{m,G}^W$, which is proportional to the reciprocal gas pressure Eq. (10). Due to thermal expansion of the gaseous phase, a gas pressure

gradient with higher pressures at the heater forms, which cannot form in the TH²M-R model. With the definition of $x_{m,G}^W$ Eq. (10), this gas pressure gradient weakens the gradient of $x_{m,G}^W$, consequently weakening the vapour diffusion away from the heater. However, this also leads to a higher net-inflow of the water component from the Opalinus Clay into the bentonite in the TH²M model (advective transport of the water component in the liquid phase towards the heater and diffusive

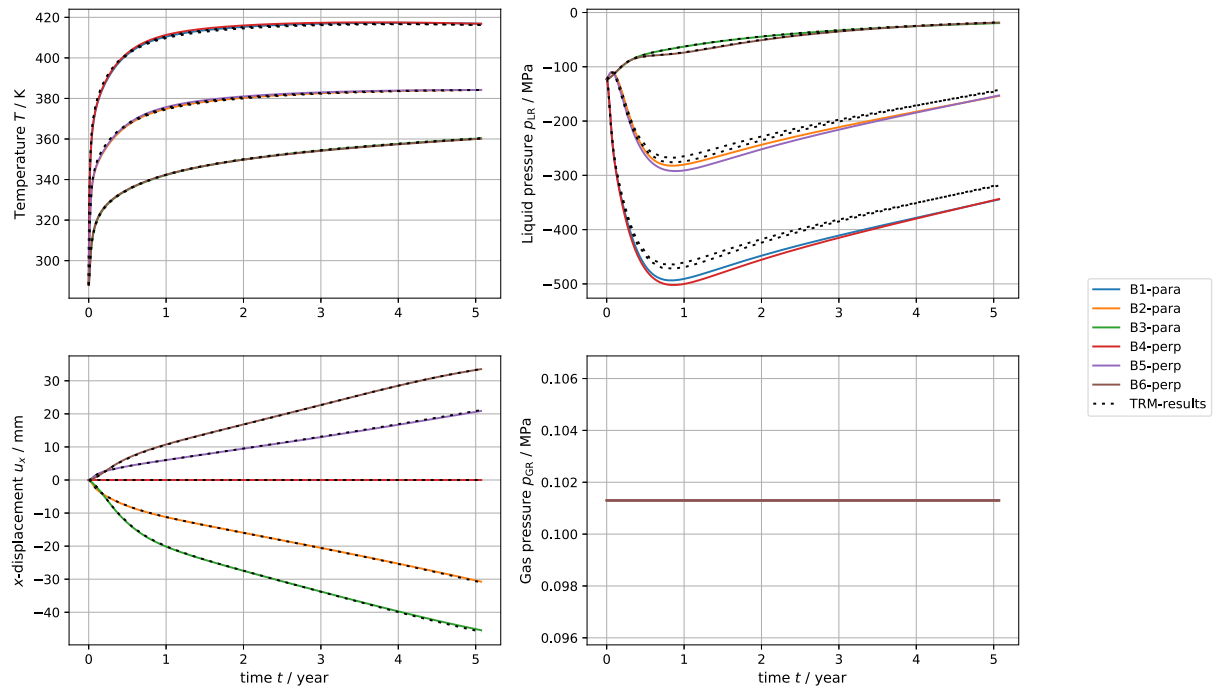


Fig. 5. Comparison of results from the bentonite backfill obtained by the TRM (black dashed) and TH²M-R (coloured solid) models in the heterogeneous case with initially unsaturated material (bentonite backfill around the heater) and fully saturated material (host rock). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

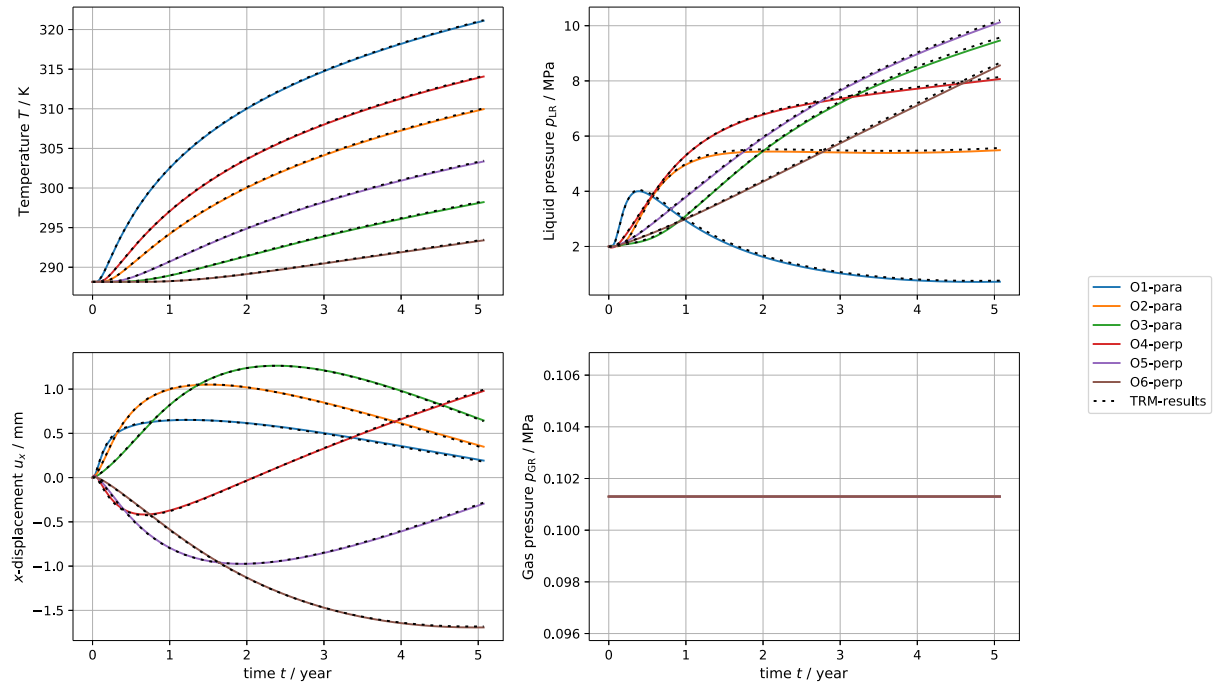


Fig. 6. Comparison of results from the Opalinus Clay obtained by the TRM (black dashed) and TH²M-R (coloured solid) models in the heterogeneous case with initially partially saturated material (bentonite backfill around the heater) and fully saturated material (host rock). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

transport of the water component in the gaseous phase away from the heater are opposing forces in this example). This interpretation is supported by the fact that observation points in the bedding plane of the Opalinus Clay experience higher saturation and thus less dry-out (cf. liquid pressures in Fig. 7) than their counterparts perpendicular to the bedding plane which is a result of the anisotropic permeability of the host rock. Lastly, the initial sharp increase in gas pressure both in

the bentonite (Fig. 7) and the host rock (Fig. 8) is probably mainly due to thermal expansion as a result of heating.

5. Discussion

It is difficult to quantify the impact of a variable which was neglected in the underlying physical assumptions constituting a numerical model. This is the case for the thermo-hydro-mechanically coupled

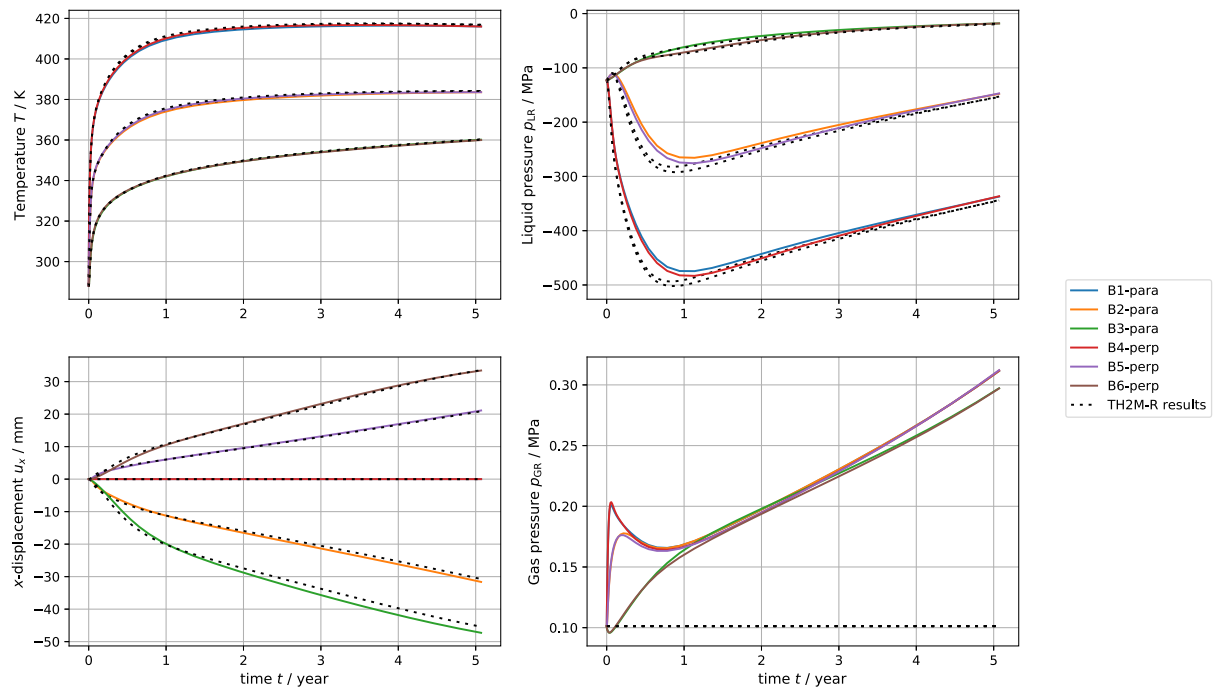


Fig. 7. Comparison of results obtained in the bentonite by the TH²M-R model with a fixed gas pressure (black dashed) and the TH²M model with a free gaseous phase (coloured solid). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

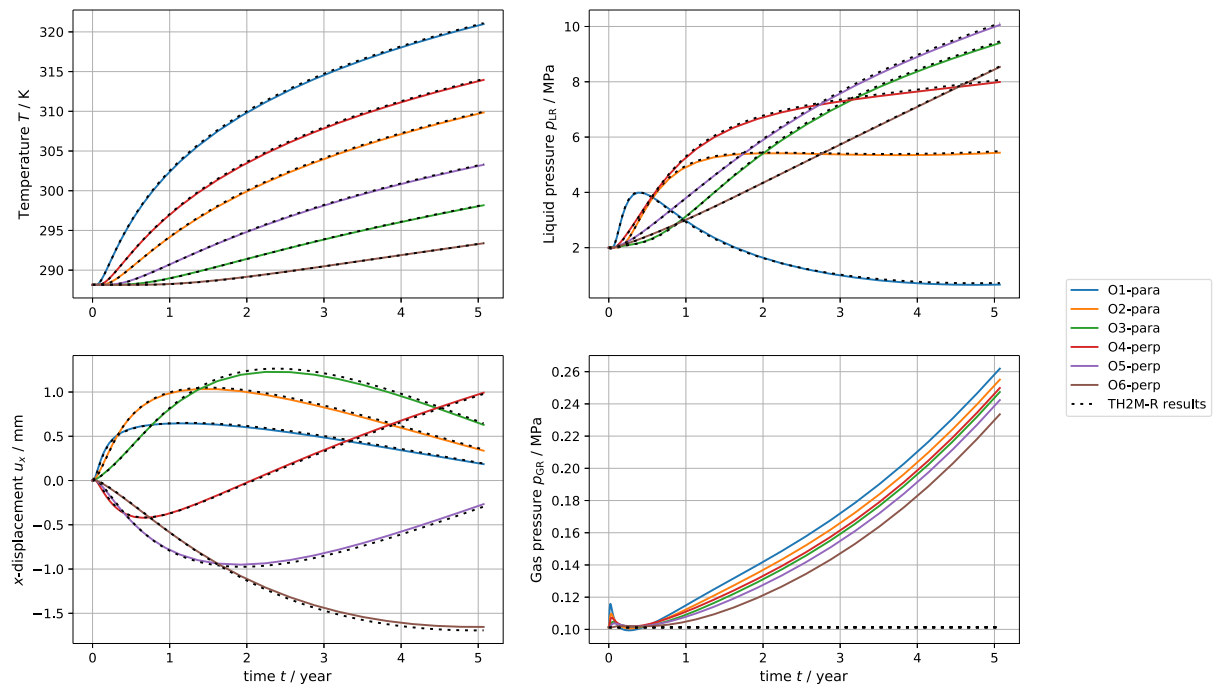


Fig. 8. Comparison of results obtained in the Opalinus Clay by the TH²M-R model with a fixed gas pressure (black dashed) and the TH²M model with a free gaseous phase (coloured solid). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

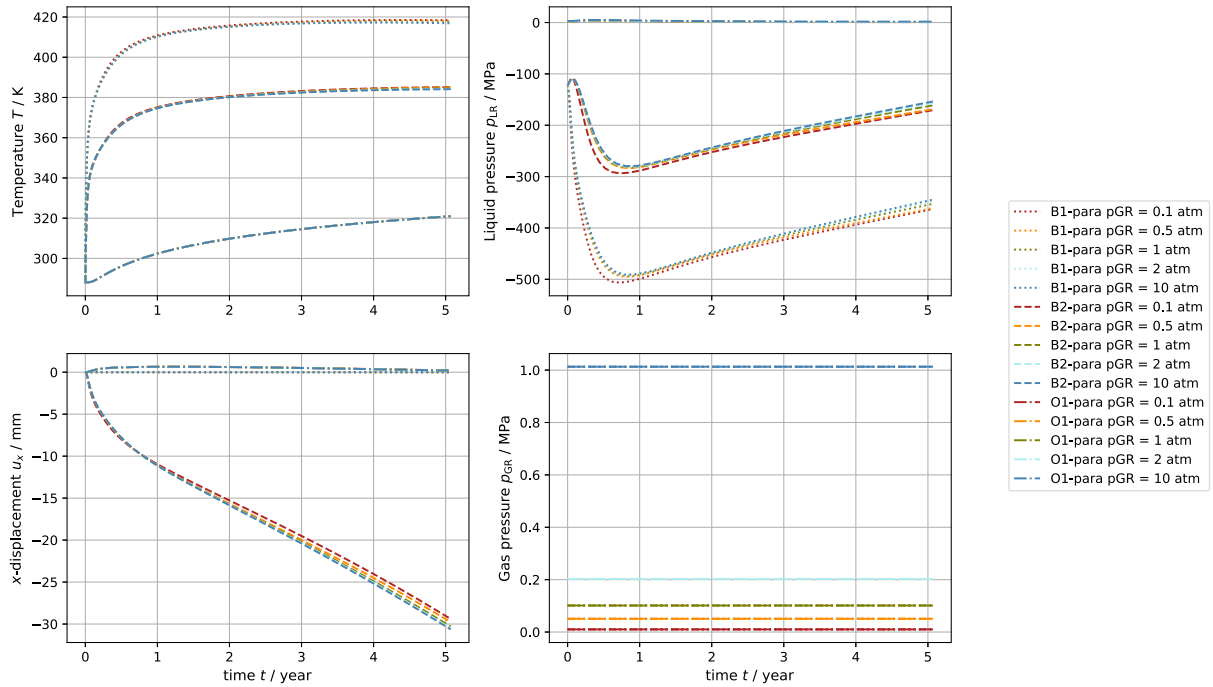


Fig. 9. Sensitivity of results obtained by the TH²M-R model with regard to the constant gas pressure given for the Richards case.

TRM implementation when it comes to the impact of a free gas pressure. A suitable way to confirm if the application of the Richards approximation is justified in a given situation is the comparison of TRM results with results obtained by the more complex TH²M model, which does not disregard the effects of a gaseous phase in the first place. Differences between the governing equations of the TRM and TH²M implementations due to the presence of a non-aqueous component or phase can be identified in every one of the coupled processes.

When comparing the governing equations of the TH²M-R and TRM models, where gas pressure is assumed to be constant, still several differences arise from this constraint:

- Because of $p_{\text{cap}} = p_{\text{GR}} - p_{\text{LR}}$, an initial capillary pressure has to be chosen for TH²M in the Opalinus Clay which is *different* from the equivalent TRM initial liquid pressure – depending on the choice for the gas pressure – in order to match initial liquid pressures computed by the TH²M model and given for the TRM model. In the unsaturated regions however, the capillary pressure is set to match the given initial saturation. It is therefore *equivalent* to the initial liquid pressure set in TRM and is *not* offset by the initial gas pressure. At the interface of saturated and unsaturated regions, this changes liquid pressure gradients as a function of initial gas pressure.
- A residual p_{GR} has implications for the momentum balance. A constant gas pressure multiplied by a non-constant gas saturation leads to a non-zero divergence term and therefore causes a force density which may become significant at high gas pressures.
- The residual gas pressure impacts the diffusion as shown in Eq. (37) because of the dependency of the gaseous phase mixture on the gas pressure. It should be noted that the dependence of the diffusion on the molar mass mixtures (cf. Eqs. (37) and (9)) could be different, depending on the gaseous phase composition.

The effects listed above are illustrated by comparing the results of several TH²M-R models using 0.1 atm, 0.5 atm, 1 atm, 2 atm and 10 atm as constant values for the gas pressure in Fig. 9. As discussed in

Section 4.3, a lower gas pressure leads to a stronger diffusive transport of the water component, thus resulting in a stronger dry-out in the near-field of the heater. Although the simulations with the lowest gas pressure predict the lowest liquid pressures at all times (top right in Fig. 9), the magnitude of the x-displacement is initially largest in the low-gas-pressure simulations and at later stages largest in the high-gas-pressure simulations. These differences are small, but they nevertheless illustrate the impact of a non-zero gas pressure on the momentum balance. However, the impact of a freely evolving gas pressure on the model behaviour is less subtle. Among the prominent couplings between the gas transport and other processes are:

- Because of $p_{\text{LR}} = p_{\text{GR}} - p_{\text{cap}}$, changes in gas pressure affect the liquid pressure and consequently advective flux in the liquid phase.
- A non-constant p_{GR} can have significant implications for the momentum balance. However, since the present simulations do not consider high gas pressures due to gas generation, this impact is within reasonable limits and the use of the Richards approximation remains justified.
- The behaviour of the gaseous phase heavily impacts the transport of the components. The water component can be transported advectively and diffusively in both phases and thus, energy in the form of latent heat or stored as pressure work can be transported through the domain.
- The gaseous phase can displace liquid and change bulk properties.

As shown in this work, some differences arise from the presence of a gaseous phase with constant pressure and when a free gas pressure is considered, the differences become more notable. However, they stay within limits, and the general system behaviour is predicted in a very similar way by both models. As the TH²M model is the more complex model, it can be used to quantify the impact of constitutive choices underlying the TRM model. In the present case, the TRM model performs well and its usage is justified. However, it is easily conceivable that higher gas pressures could occur in a HLW repository and the

results of this work clearly indicate that the over-all behaviour of the system makes it necessary to consider a free gas pressure including phase transitions of both components in both phases. It is at that point that the limits of the TRM model are reached, as the model assumptions are over-stretched (cf. Fig. 1).

In general, the hydraulic, thermal and mechanical processes predicted by the TH²M and TRM models for the particular example chosen in this work show very good agreement.¹⁹ found that thermal results of Richards-flow and two-phase-flow models agreed well in a heat pipe test, but agreement of hydraulic results was highly dependent on the intrinsic permeability. However, the bespoke heat pipe test was carried out on a scale of centimetres rather than a meter scale and under these circumstances, a pressurisation of the gas phase might have a stronger influence than in the test case discussed in the present work, where much larger volumes are available in the bentonite. In addition, the aim of Ref. 19 was to fit simulation results to measurement data, which included using permeability as a fitting parameter. As we discuss in the following paragraph, we also note a strong dependence of results and especially gaseous phase behaviour on the permeability. The aim of the present study was to leave input parameters for the TH²M and TRM models as similar as possible and to use equivalent parameters, where the models required it. As a consequence, we find a better agreement of model predictions, but it has to be noted that this finding is only valid for the present test case and has to be re-evaluated for other applications (especially if geometry, material parameters and boundary conditions deviate from the present case).

The final thought is on the gas mobility and the two-phase-model in zones of full saturation. In the unsaturated bentonite, gas permeability has to be assumed to be up to 1000 times higher than the given intrinsic permeability in order to suppress the formation of strong gas pressure gradients. Due to effects such as the Klinkenberg effect or Knudsen diffusion, gas permeability may be higher by orders of magnitude in highly porous, low saturation materials such as in the cases discussed in this work.⁴⁷ For unsaturated Opalinus Clay, it has been shown that the gas permeability can range in the order of $1 \times 10^{-17} \text{ m}^2$ or higher, which is up to five orders of magnitude higher than the intrinsic permeability of the host rock used in this study.⁴⁸ There is a need to further study this effect as the impact on gas transport could be significant. In the fully saturated zones, however, the relative permeability of the gaseous phase is modelled effectively equal to zero, if a van Genuchten relative permeability is used. As a consequence, the only mode of transport for the non-aqueous component is diffusion in the residual gas volume, which is independent of relative permeability, and the results are low gas pressures. For future works with the TH²M implementation, saturated regions should be treated by allowing for the non-aqueous component to be dissolved in the liquid phase in order to represent both diffusion and advection dominated gas transport regimes.^{49–51}

6. Conclusions

The present work illustrates that both the TRM and TH²M implementations are well capable of simulating the complex process of dry-out and subsequent resaturation of bentonite as well as the thermal consolidation of the host rock and thermal pressurisation of pore fluids around a heat-releasing nuclear waste canister. In simplified conditions, such as a fully liquid saturated medium, the governing equations collapse as any terms related to unsaturated processes vanish and end up to be virtually identical. Moreover, the TH²M model is capable of representing a more complex process with one additional degree of freedom (corresponding to the independent gas pressure). However, the assumptions related to the Richards approximation can be explicitly considered in the governing equations of the TH²M model, which collapse into the simpler TH²M-R model, which is similar to the TRM implementation. Furthermore, it was shown that the two formulations approach each other for high liquid saturation and small vapour densities. It was illustrated that – although utilising different

formulations – the vapour diffusion in the TRM and TH²M models can be re-formulated in similar ways yielding mostly similar results, especially in the case of constant gas pressure. It could furthermore be shown that for the FE-experiment example, the model is not particularly sensitive to the constant gas pressure chosen, and comparing the TRM and TH²M results in their respective full complexity, the results reflect the same system behaviour. This is a good indicator that the TRM model is well suited for investigations concerning modelling experiments like the FE-experiment. On the basis of the PDEs of the more complex TH²M, several potential impact possibilities of a non-constant gas pressure on the THM-coupled system were illustrated, which all could give rise to conceptual model uncertainty when utilising the Richards approximation. However, the presented test cases illustrate that in the case of the FE-experiment, these effects and thus the related uncertainty remain within reasonable limits. In general, it seems to be a suitable workflow to use the TH²M model as a first reference for any modelling problem in order to determine the consequences associated with invoking the Richards approximation. After showing that a freely evolving gas pressure does not lead to significantly different results and that there is no significant gas pressure build-up to be expected, one may choose to progress the analyses using the TRM implementation, which is after all the faster one due to the reduced degree of freedoms.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Evaluation of the TH²M water component mass balance, expansion of the density derivatives $\frac{d\rho_{GR}^W}{dt}$, $\frac{d\rho_{LR}^W}{dt}$ and reordering with respect to the driving forces.

We start with Eq. (2) with $\zeta \equiv W$:

$$\begin{aligned} 0 = & \rho_{FR}^W (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{GR}}{dt} - \rho_{FR}^W (\alpha_B - \phi) \beta_{p,SR} S_L \frac{dp_{cap}}{dt} \\ & - \rho_{FR}^W (\alpha_B - \phi) \text{tr} \left(\alpha_{T,SR} \right) \frac{dT}{dt} \\ & + \phi \left[(1 - S_L) \frac{d\rho_{GR}^W}{dt} + S_L \frac{d\rho_{LR}^W}{dt} \right] \\ & + [\phi(\rho_{LR}^W - \rho_{GR}^W) - \rho_{FR}^W p_{cap} (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt} \\ & + \rho_{FR}^W \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div} (\mathbf{A}^W + \mathbf{J}^W). \end{aligned} \quad (39)$$

The density derivative terms (first terms second line in above equation) can be expressed by

$$\begin{aligned} \frac{d\rho_{GR}^W}{dt} &= \frac{d\rho_{GR}^W}{dt} = \frac{\partial \rho_{GR}^W}{\partial p_{GR}} \frac{dp_{GR}}{dt} + \frac{\partial \rho_{GR}^W}{\partial p_{cap}} \frac{dp_{cap}}{dt} + \frac{\partial \rho_{GR}^W}{\partial T} \frac{dT}{dt} \\ \text{and} \\ \frac{d\rho_{LR}^W}{dt} &= \frac{\partial \rho_{LR}^W}{\partial p_{LR}} \frac{dp_{LR}}{dt} + \frac{\partial \rho_{LR}^W}{\partial T} \frac{dT}{dt} = \frac{\partial \rho_{LR}^W}{\partial p_{LR}} \frac{d(p_{GR} - p_{cap})}{dt} + \frac{\partial \rho_{LR}^W}{\partial T} \frac{dT}{dt} \\ &= \frac{\partial \rho_{LR}^W}{\partial p_{LR}} \frac{dp_{GR}}{dt} - \frac{\partial \rho_{LR}^W}{\partial p_{LR}} \frac{dp_{cap}}{dt} + \frac{\partial \rho_{LR}^W}{\partial T} \frac{dT}{dt} \\ &= \rho_{LR}^W \beta_{p,LR} \frac{dp_{GR}}{dt} - \rho_{LR}^W \beta_{p,LR} \frac{dp_{cap}}{dt} - \rho_{LR}^W \beta_{T,LR} \frac{dT}{dt} \end{aligned} \quad (40)$$

and inserted in Eq. (39), yielding:

$$\begin{aligned} 0 = & \rho_{FR}^W (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{GR}}{dt} - \rho_{FR}^W (\alpha_B - \phi) \beta_{p,SR} S_L \frac{dp_{cap}}{dt} \\ & - \rho_{FR}^W (\alpha_B - \phi) \text{tr} \left(\alpha_{T,SR} \right) \frac{dT}{dt} \\ & + \phi \left[(1 - S_L) \left(\frac{\partial \rho_{GR}^W}{\partial p_{GR}} \frac{dp_{GR}}{dt} + \frac{\partial \rho_{GR}^W}{\partial p_{cap}} \frac{dp_{cap}}{dt} + \frac{\partial \rho_{GR}^W}{\partial T} \frac{dT}{dt} \right) \right. \\ & \left. + S_L \left(\rho_{LR}^W \beta_{p,LR} \frac{dp_{GR}}{dt} - \rho_{LR}^W \beta_{p,LR} \frac{dp_{cap}}{dt} - \rho_{LR}^W \beta_{T,LR} \frac{dT}{dt} \right) \right] \\ & + \rho_{FR}^W \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div} (\mathbf{A}^W + \mathbf{J}^W) \\ & + [\phi(\rho_{LR}^W - \rho_{GR}^W) - \rho_{FR}^W p_{cap} (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}. \end{aligned} \quad (41)$$

We can now apply the definition of the effective component density in all phases where we note that due to no gas dissolution $\rho_{LR}^W = \rho_{LR}$. Thus, we can substitute $\rho_{FR}^W = (1 - S_L) \rho_{GR}^W + S_L \rho_{LR}$ and arrive at:

$$\begin{aligned} 0 = & \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{GR}} + \phi S_L \rho_{LR} \beta_{p,LR} \right) \frac{dp_{GR}}{dt} \\ & + ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{cap}}{dt} \\ & + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial p_{cap}} - \phi S_L \rho_{LR} \beta_{p,LR} \right) \frac{dp_{cap}}{dt} \\ & - ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) (\alpha_B - \phi) \beta_{p,SR} S_L \frac{dp_{cap}}{dt} \\ & + \left(\phi(1 - S_L) \frac{\partial \rho_{GR}^W}{\partial T} - \phi S_L \rho_{LR} \beta_{T,LR} - ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) \right) \frac{dT}{dt} \\ & \times (\alpha_B - \phi) \text{tr} \left(\alpha_{T,SR} \right) \\ & + ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) \alpha_B \text{div} \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div} (\mathbf{A}^W + \mathbf{J}^W) \\ & + [\phi(\rho_{LR} - \rho_{GR}^W) - ((1 - S_L) \rho_{GR}^W + S_L \rho_{LR}) p_{cap} (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt}. \end{aligned} \quad (42)$$

Appendix B. Expression of the vapour diffusion in TH²M in terms of vapour density

In the TH²M implementation, the diffusion of the water component in the gaseous phase is expressed in terms of the mass fraction $x_{m,G}^W$, which is calculated based on the vapour pressure. In the TRM implementation, the vapour diffusion is expressed in terms of vapour density. In order to compare the two formulations, one may convert the mass fraction formulation as follows. Firstly, we express the diffusive flux given for the TH²M-R model in Eq. (7) by noting that $\text{grad } p_{GR} = 0$ in the TH²M-R model and noting that the differential of the mass fraction $x_{m,G}^W$ can be expressed by $dx_{m,G}^W = \frac{M_W M_C}{M_G^2} dx_{n,G}^W$:

$$\begin{aligned} \mathbf{J}_G^W &= -\rho_{GR} D_G^W \left(\frac{\partial x_{m,G}^W}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial x_{m,G}^W}{\partial T} \text{grad } T \right) \\ &= -\rho_{GR} \frac{M_W M_C}{M_G^2} D_G^W \left(\frac{\partial x_{n,G}^W}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial x_{n,G}^W}{\partial T} \text{grad } T \right). \end{aligned} \quad (43)$$

We insert Eq. (10) and consider that – as long as it is constant – p_{GR} can be moved in front of the derivatives acting as a scalar:

$$\mathbf{J}_G^W = -\rho_{GR} \frac{1}{p_{GR}} \frac{M_W M_C}{M_G^2} D_G^W \left(\frac{\partial p_{GR}^W}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial p_{GR}^W}{\partial T} \text{grad } T \right). \quad (44)$$

Next, we relate vapour pressure to vapour density using the ideal gas law $p_{GR}^W = \rho_{GR}^W \frac{RT}{M_w}$, bringing us to:

$$\begin{aligned} \mathbf{J}_G^W &= -\rho_{GR} \frac{1}{p_{GR}} \frac{M_W M_C}{M_G^2} D_G^W \left(\frac{\partial \rho_{GR}^W \frac{RT}{M_w}}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial \rho_{GR}^W \frac{RT}{M_w}}{\partial T} \text{grad } T \right) \\ &= -\rho_{GR} \frac{1}{p_{GR}} \frac{M_W M_C}{M_G^2} D_G^W \left(\frac{RT}{M_w} \frac{\partial \rho_{GR}^W}{\partial p_{cap}} \text{grad } p_{cap} \right. \\ &\quad \left. + \left(\frac{R \rho_{GR}^W}{M_w} + \frac{RT}{M_w} \frac{\partial \rho_{GR}^W}{\partial T} \right) \text{grad } T \right). \end{aligned} \quad (45)$$

We collect the $\frac{RT}{M_w}$ terms outside the brackets, where M_w gets cancelled out:

$$\begin{aligned} \mathbf{J}_G^W &= -\underbrace{\frac{\rho_{GR} RT}{M_G}}_{=p_{GR}} \frac{1}{p_{GR}} \frac{M_C}{M_G} D_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{cap}} \text{grad } p_{cap} \right. \\ &\quad \left. + \left(\frac{\rho_{GR}^W}{T} + \frac{\partial \rho_{GR}^W}{\partial T} \right) \text{grad } T \right) \\ &= -\frac{M_C}{M_G} D_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{cap}} \text{grad } p_{cap} + \left(\frac{\rho_{GR}^W}{T} + \frac{\partial \rho_{GR}^W}{\partial T} \right) \text{grad } T \right). \end{aligned} \quad (46)$$

This equation expresses the vapour diffusion of the TH²M-R model in terms of vapour density gradients like it is found in Eq. (22) in the TRM implementation.

Appendix C. Overview of symbols

See Table 10.

Appendix D. Comparison results obtained by TRM and TH²M models

See Figs. 10 and 11.

Table 10

Explanation of symbols. L (unit of length), t (unit of time), T (unit of temperature), N (unit of amount of substance) and M (unit of mass) can represent any consistent set of units.

Symbol	Description	Dimension	Unit in use
ϕ_α	Property of phase $\alpha \in \{G, L, S\}$	–	–
ϕ_ζ	Property of component $\zeta \in \{W, C\}$	–	–
$\frac{d\phi}{dt}$	Material time derivative of the solid	t^{-1}	s^{-1}
$\beta_{p,SR}$	Solid compressibility	$LM^{-1}t^2$	Pa^{-1}
$\beta_{p,LR}$	Liquid compressibility	$LM^{-1}t^2$	Pa^{-1}
$\beta_{c,LR}$	Liquid expansivity due to gas dissolution	L^3N^{-1}	$m^3 mol^{-1}$
α_B	Biot coefficient	–	–
$\alpha_{T,SR}$	Solid thermal expansion tensor	T^{-1}	K^{-1}
A^ζ	Accumulative advective mass flux of ζ	$Mt^{-1}L^{-2}$	$kg s^{-1} m^{-2}$
A^ζ_α	Advective mass flux of ζ in α	$Mt^{-1}L^{-2}$	$kg s^{-1} m^{-2}$
A_α	Advective mass flux of phase α	$Mt^{-1}L^{-2}$	$kg s^{-1} m^{-2}$
C	Solid stiffness matrix	$ML^{-1}t^{-2}$	Pa^{-1}
c^C_L	Concentration of dissolved gas	NL^{-3}	$mol m^{-3}$
c_{pa}	Specific heat capacity of phase α	$L^2t^{-2}T^{-1}$	$J K^{-1} kg^{-1}$
$c_{p,vap}$	Specific heat capacity of water vapour	$L^2t^{-2}T^{-1}$	$J K^{-1} kg^{-1}$
D^ζ_α	Water vapour diffusion coefficient	L^2t^{-1}	$m^2 s^{-1}$
d^ζ_α	Diffusion velocity of ζ in α	Lt^{-1}	$m s^{-1}$
g	Gravitational acceleration acting on α	Lt^{-2}	$m s^{-2}$
H^C	Henry coefficient	Nt^2T^{-2}	$mol m^{-3} Pa^{-1}$
h^ζ, h_α	Specific enthalpy of component ζ or phase α	L^2t^{-2}	$J kg^{-1}$
J^ζ	Accumulative diffusive mass flux of ζ	$Mt^{-1}L^{-2}$	$kg s^{-1} m^{-2}$
J^ζ_α	Diffusive mass flux of ζ in α	$Mt^{-1}L^{-2}$	$kg s^{-1} m^{-2}$
k^{rel}_α	Relative permeability of α	–	–
k_S	Intrinsic permeability	L^2	m^2
L_0	Water latent heat	ML^2t^{-2}	$J m^{-3}$
λ_{eff}	Bulk thermal conductivity	$MLt^{-3}T^{-1}$	$W m^{-1} K$
M_ζ	Molar mass of ζ	MN^{-1}	$kg mol^{-1}$
M_G	Molar mass of gas mixture	MN^{-1}	$kg mol^{-1}$
μ_{aR}	Phase viscosity	$ML^{-1}t^{-1}$	$Pa s$
ϕ	Medium porosity	–	–
p_{cap}	Capillary pressure	$ML^{-1}t^{-2}$	Pa
p^C_{GR}	non-aqueous component partial pressure	$ML^{-1}t^{-2}$	Pa
p_{GR}, p_{LR}	Gaseous and liquid phase pressure	$ML^{-1}t^{-2}$	Pa
p_{SR}	Solid pressure	$ML^{-1}t^{-2}$	Pa
p^W_{GR}	Vapour pressure	$ML^{-1}t^{-2}$	Pa
p^W_{vap}	Saturated vapour pressure	$ML^{-1}t^{-2}$	Pa
ρ_{aR}	Phase density	ML^{-3}	$kg m^{-3}$
ρ^c_{FR}	Effective component density	ML^{-3}	$kg m^{-3}$
ρ^c_{GR}, ρ^c_{LR}	Density of ζ in gaseous and liquid phase	ML^{-3}	$kg m^{-3}$
ρ_{GR}, ρ_{LR}	Gaseous and liquid phase density	ML^{-3}	$kg m^{-3}$
$\rho_{LR,ref}$	Reference liquid phase density	ML^{-3}	$kg m^{-3}$
ρ^W_{GR}	Vapour density	ML^{-3}	$kg m^{-3}$
R	Universal gas constant	$ML^2t^{-2}N^{-1}T^{-1}$	$J K^{-1} mol^{-1}$
S_L	Liquid saturation	–	–
$S_{L,max}, S_{L,res}$	Maximum and residual liquid saturation	–	–
S_e, S_{entry}	Effective liq. sat. and saturation at gas entry	–	–
σ'	Effective stress tensor	$ML^{-1}t^{-2}$	Pa
T	Temperature	T	K
T_{ref}	Reference temperature	T	K
θ_{vap}	Volumetric vapour content	–	–
u_α	Specific internal energy of phase α	L^2t^{-2}	$J kg^{-1}$
u_S	Solid displacement vector	L	m
$\tilde{w}_{aS}$	Darcy velocity w.r.t solid	Lt^{-1}	$m s^{-1}$
$x^c_{m,\alpha}$	Mass fraction of component ζ in phase α	–	–
$x^c_{n,\alpha}$	Molar fraction of component ζ in phase α	–	–
χ	Bishop's coefficient	–	–

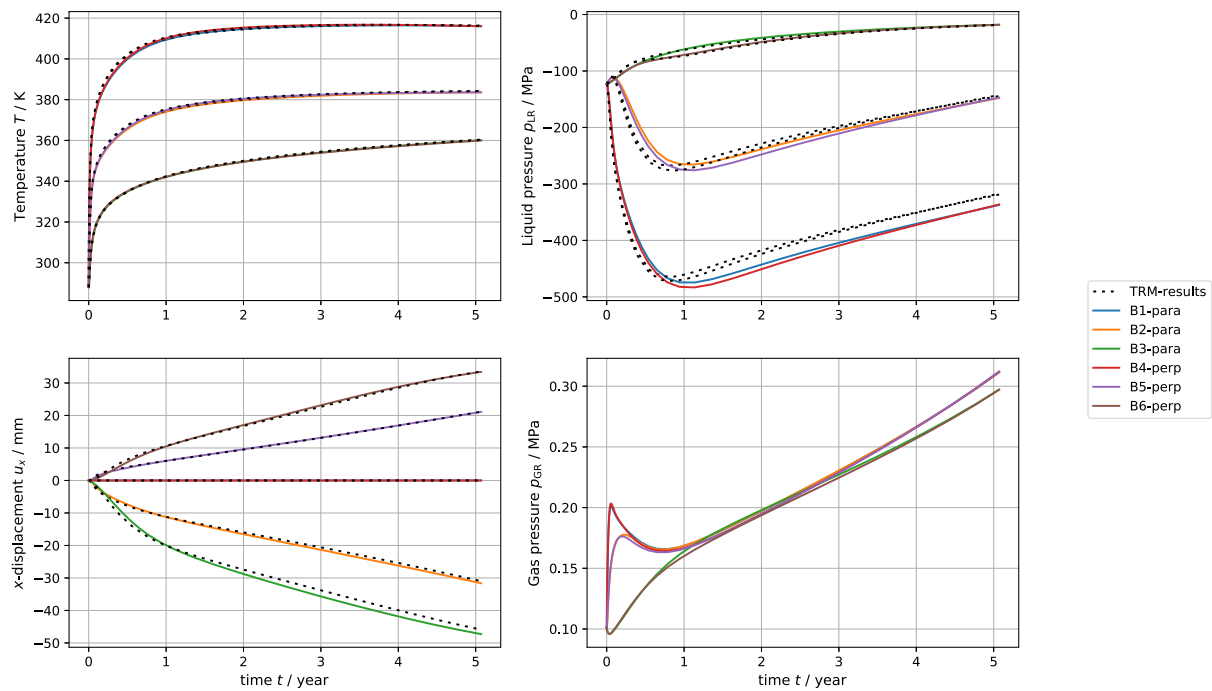


Fig. 10. Comparison of results obtained for the points in the bentonite by the TRM and TH²M models in their respective full complexity.

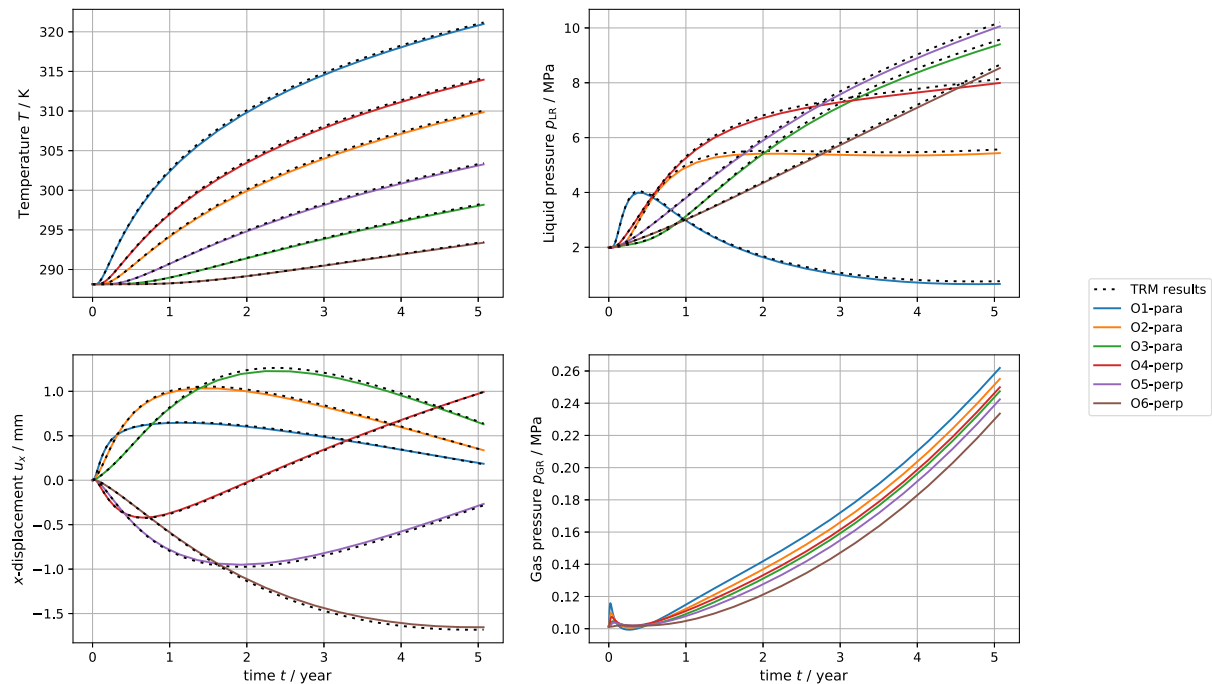


Fig. 11. Comparison of results obtained for the points in the Opalinus Clay by the TRM and TH²M models in their respective full complexity.

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